

Retail Radar

Jul 29 – Momentum Unwind, Cutting Tech Losses

Retail Trading Activity Overview

- **Sentiment deteriorated further this week** as retail flows meaningfully weakened, reaching their 0.8%ile vs 1y history; this time, retail investors **aggressively pulled out of single names** (3.2%ile, with 4 consecutive days of single-stock selling, Figure 10), while only modestly adding exposure to ETFs (1.6%ile).
- **Within ETFs**, Tech saw significant outflows—the second-largest on record, behind only the week ending June 10th, Figure 5, driven primarily by SOXL (-1.2z, Figure 6). Retail investors also cut exposure to crypto ETFs, led by BITO (-4.9z).
- **Single-stock selling was overwhelmingly concentrated in Tech.** Hardware names took the brunt of the selling, posting their largest weekly selling on record, Figure 7. SNDK was the biggest driver (-3.2z), followed by AAPL. In broader Semis, MU moved into the most-sold list among retail investors—just a few weeks after consistently featuring alongside SNDK among their favorites, Figure 8. In line with this, our Crowding analysis that captures broader market positioning quantifies the sharp Semis de-crowding, down to the 91.8%ile from the 99%ile, see [report](#) and [report](#), Figure 11. In contrast, **the Mag 7 + SPCX saw positive flows overall**: NVDA, GOOGL, SPCX and TSLA were amongst the top 5 this week, alongside SKHY. Internationally, Korea has undergone an intense period of de-leveraging since mid-June, with the positioning setup now appearing attractive on balance according to our analysts, see [report](#).
- **The sharp deterioration in retail YTD P&L (Figure 9) may help explain why retail investors have been quick to cut Tech exposure** as the Semis-led Momentum Unwind has weighed on portfolios (see Factor Crowdings and Rotations).
- As we head into peak-Earnings, investors are focusing on the major announcements slated for this week. Retail participation has been limited so far, underscoring a deterioration in sentiment, see Single Stock Stories with Elevated Retail Activity:
 - Retail investors trimmed META, MSFT and QCOM on Wednesday before their announcements (-0.6z, 0.8z, and -0.2z, respectively). **MSFT's** Cloud unit posted a larger-than-expected increase in sales at +43% y/y, with shares rising ~3% in post-market trading. Capex increased +70% y/y to \$41B during the quarter. **META** narrowed its 2026 capex guidance to \$130-145B, lifting the low-end from \$125B. Revenue for the quarter came in line with expectations, but next quarter's outlook was lackluster. Shares fell as much as ~10% in post-market trading after the results were published. **QCOM's** earnings missed expectations, but top line was better than expected. Additionally, the company expects non-handset revenue growth to accelerate to 60% in fiscal year 2027, as it tries to reduce its dependence on phones with a new push into the data center market.
 - Retail investors trimmed AAPL and AMZN on Wednesday (-0.7z and -0.6z, resp.). **AMZN** reports earnings tomorrow AMC, with our analysts

Equity Strategy and Quantitative Research

Arun Jain ^{AC}

(1-212) 622-9454
arun.p.jain@jpmorgan.com
J.P. Morgan Securities LLC

Shizuka Suga, CFA ^{AC}

(1-212) 622-2134
shizuka.suga@jpmorgan.com
J.P. Morgan Securities LLC

Ana Pous Avila

(1-212) 622-0496
ana.pousavila@jpmorgan.com
J.P. Morgan Securities LLC

William Matheson

(1-212) 622-9538
william.matheson@jpmchase.com
J.P. Morgan Securities LLC

Khuram Chaudhry

(44-20) 7134-6297
khuram.chaudhry@jpmorgan.com
J.P. Morgan Securities plc

Bhupinder Singh ^{AC}

(1-212) 622-9812
bhupinder.singh@jpmorgan.com
J.P. Morgan Securities LLC

Dubravko Lakos-Bujas ^{AC}

(1-212) 622-3601
dubravko.lakos-bujas@jpmorgan.com
J.P. Morgan Securities LLC

Past Issues:

[7/29 - Momentum Unwind Accelerates Amid Earnings](#)
[7/22 - Momentum Cooldown, Hyperscaler Earnings](#)
[7/15 - Trading Earnings as Flows Cool](#)
[7/8 - Split-the-Dip as Momentum Cools](#)
[7/1 - Same Playbook, Buying the Dip](#)
[6/24 - Remembers the Dip, MU Serves Right](#)
[6/17 - Tech Craze Spillover to Communications](#)
[6/10 - Tech Selling amidst Drawdown, World Cup](#)
[6/3 - MEME Earnings, Trading around IPOs](#)
[5/27 - Theme Stack: AI, Memory, Space, Quantum](#)
[5/13 - Software vs Semis Positioning](#)
[5/6 - Memory and AI, the Catch-Up Trade](#)
[4/29 - Peak Earnings, Peak Mag-7](#)
[4/22 - Towards a MEME Revival?](#)
[4/15 - Back to Average](#)
[4/8 - Not Buying Despite the Ceasefire](#)
[4/1 - In Defensive Mode](#)
[3/25 - Selling Rips](#)
[3/18 - From FOMO to FOHO: Fear Of Higher Oil](#)

See page 23 for analyst certification and important disclosures.

expecting AWS growth acceleration of low-mid 30%'s, higher capex guidance and retail margin expansion ([link](#)). AAPL also reports tomorrow, with price elasticity the key watchpoint after ~20% average price hikes across iPads and Macs—moves that have also lifted expectations for iPhone 18 pricing. While investors debate near-term consumer pushback amid higher memory costs, our analysts see multiple offsets that should leave revenue and earnings more resilient than feared ([link](#)).

[3/11 - Buying Oil, Selling Energy](#)
[3/4 - Cautiously Optimistic](#)
[2/25 - Balanced Optimism](#)
[2/18 - AI Software Averaging Down](#)
[2/11 - Positioning around Dislocations](#)
[2/4 - Softening Sentiment, Software vs Semis](#)
[1/28 - Earnings, Intl Equities, Precious Metals](#)
[1/21 - Unfazed through Vol Shock](#)
[1/14 - The January Effect?](#)
[1/7 - December Spree and a good start to January](#)
[12/10 - 2025 Retail Mania](#)
[12/3 - The Retail AI Trade](#)
[11/26 - Weekly Highlights](#)
[11/19 - 2025 Retail Mania: Dip-buying in AI](#)
[11/12 - Bullion to Silicon: AI Gold Rush Resumes](#)
[11/5 - Skipping the Dip, Earnings Stock Picking](#)
[10/29 - Reaching Peak-Earnings, AI, FOMC](#)
[10/22 - Sold GLD, onto New Memes and Earnings](#)
[10/15 - Trading around Tariffs, AI and Earnings](#)
[10/8 - Back after Summer Break](#)
[10/1 - Back to Stocks, Thematic ETFs, GLD Rush](#)
[9/24 - Precious Metals, Cyclical and Domestic](#)
[9/17 - FOMC, Herding in OPEN and GRAB](#)

We thank G. K. Kiruthik Srinivas and Aditya Mulye of J.P. Morgan India Private Limited for their significant contributions to this report.

Factor Crowdings and Rotations

- Amid weakening retail sentiment, a Momentum Unwind, and the recent market drawdown, we provide a few additional observations on investor positioning:
 - (1) Crowding remains high across higher-risk segments (e.g., Low Quality at 94.8%ile and Speculative Growth at 98.0%ile), with Value also elevated (88.7%ile);
 - (2) Momentum's Beta exposure is high, and Momentum–Beta correlations have declined from extremes during this unwind; and
 - (3) Implied stock correlations increased over the past week from historically low levels (COR3M <Index>), even as peak-Earnings season typically coincides with lower stock correlations. In many risk-model frameworks, higher implied correlations translate into higher implied betas and higher portfolio risk estimates, which lead systematic/quant portfolios to reduce net exposure as volatility targets and risk constraints bind.
- In combination with month-end Momentum rebalancing, these conditions are consistent with the elevated volatility observed in Momentum and higher-risk factors this week.

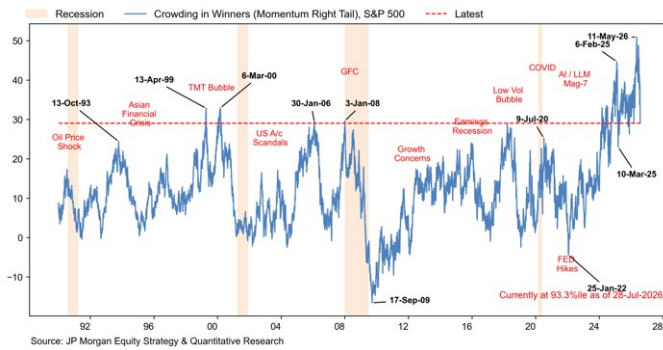
Retail Trading Activity in Numbers for the Week: Jul 23 to Jul 29

- Retail flows were \$2.2B this week, well below the 12-month average of \$6.8B/week. Retail investors continued to favor ETFs (+\$3.7B) over Single Stocks (-\$1.4B).
- Outside of **Broad Based Large Cap Equity ETFs** (+\$1.5B), retail ETF buying was influenced by **Equity Style Dividend** (+\$261M), **Equity Style Call/Put writing** (+206M), **Fixed Income – Multi Sector** (+\$172M), and **Equity Broad Based Multi Cap** (+\$166M).
- This week, in line with prior weeks, retail investors **continued to buy AI datacenters and electrification (JPAMAIDE), Mag 7, Top 30 AI/Datacenter Beneficiaries, Growth**, along with **AI Software/Product/Monetization**, and **OBBBA Immediate Expensing Beneficiaries**, see Figure 4.
- Activity in Mag7** this week: Retail investors bought NVDA (+\$601M), GOOGL/GOOG (+\$377M), TSLA (+\$121M), MSFT (+\$33M) and sold AMZN (-\$22M), META (-\$26M) and AAPL (-\$169M).
- Outside of Mag7**, retail investors favored Communications (+\$177M) and were net sellers of Tech (-\$2.2B), Industrials (-\$232M), Cons. Disc. (-\$200M) and Health Care (-\$176M).
- Top five retail stocks last week:** NVDA (+\$601M), GOOGL/GOOG (+\$377M), SPCX (+\$263M), SKHY (+\$166M), TSLA (+\$121M). **Bottom 5 retail stocks:** SNDK (-\$579M, -3.2z), MU (-\$515M, -1.9z), AMD (-\$328M, -2.2z), AAPL (-\$169M), MRVL

(-\$84M)

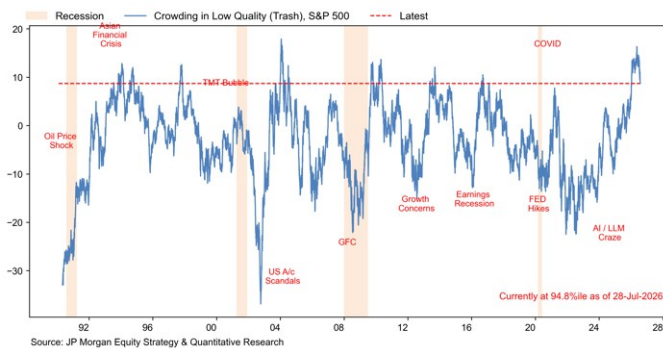
- **In options**, retail share is at highs, Figure 54. The most traded options were in the following names, echoing previous weeks: TSLA, NVDA, MU, AMZN, SNDK, META, AMD, AAPL, SPCX, and MSFT. For the most bought/sold delta and gamma names, see Retail Activity in Options. Also see Retail Investors Options Activity by Sectors.

Figure 1: Momentum Crowding



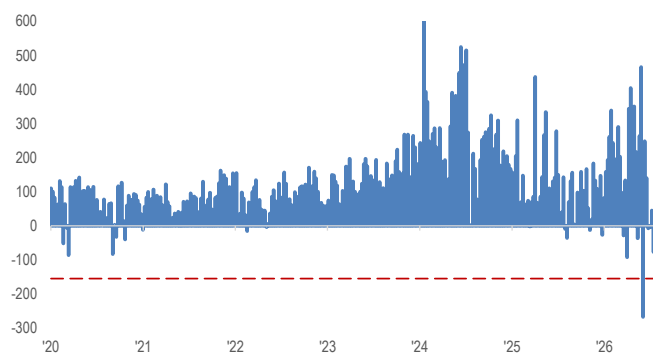
Source: J.P. Morgan Equity Strategy & Quantitative Research

Figure 3: Crowding in Low Quality



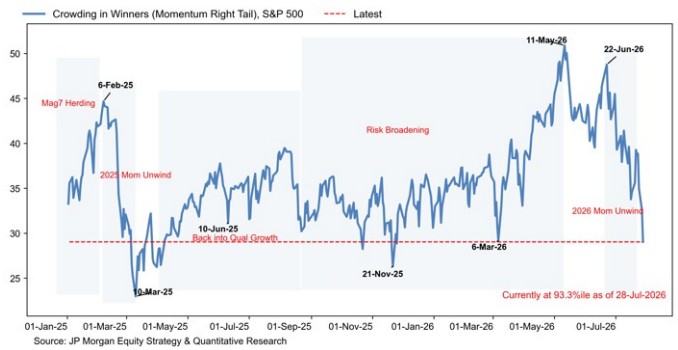
Source: J.P. Morgan Equity Strategy & Quantitative Research

Figure 5: Weekly Outflows out of Tech ETFs were the second highest on record



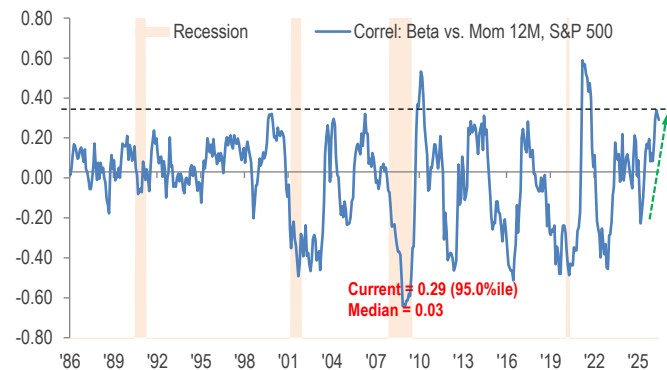
Source: J.P. Morgan Equity Strategy & Quantitative Research

Figure 2: Momentum Unwind Now vs Feb-Mar'25



Source: J.P. Morgan Equity Strategy & Quantitative Research

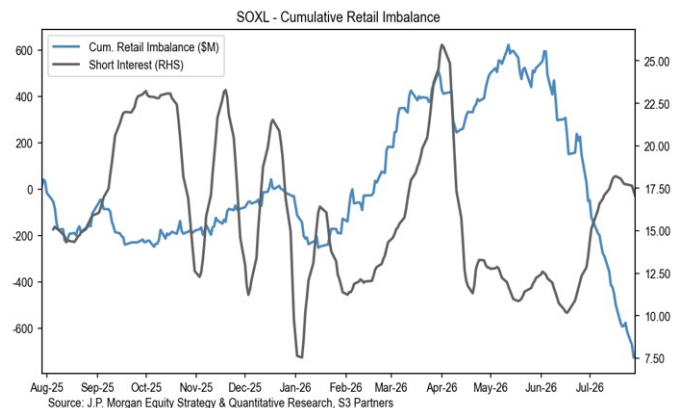
Figure 4: Momentum Beta Correlations



Source: J.P. Morgan Equity Strategy & Quantitative Research

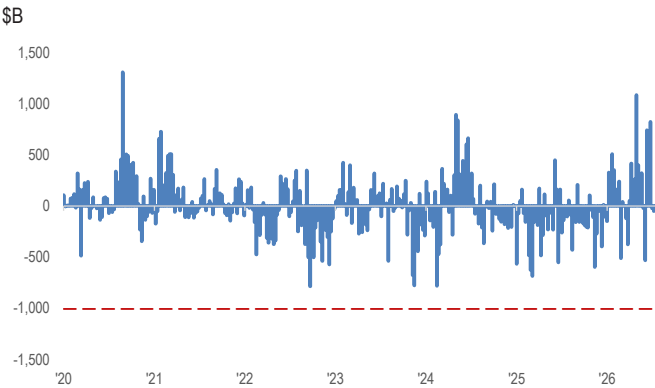
Figure 6: Cumulative Retail Imbalance in SOXL

As of Jul 29th



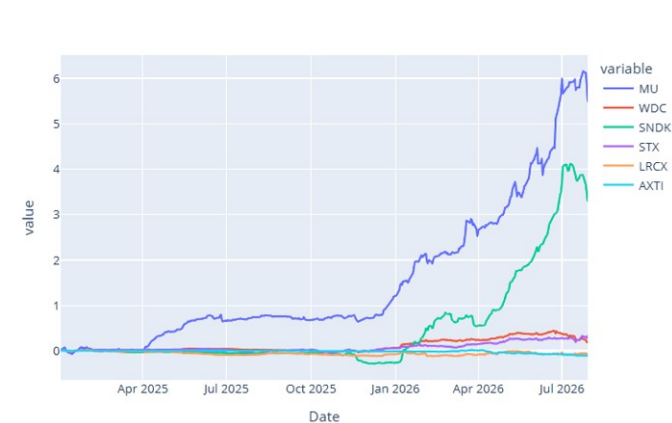
Source: J.P. Morgan Equity Strategy & Quantitative Research

Figure 7: Weekly Outflows out of Tech Hardware & Equipment stocks were the highest on record



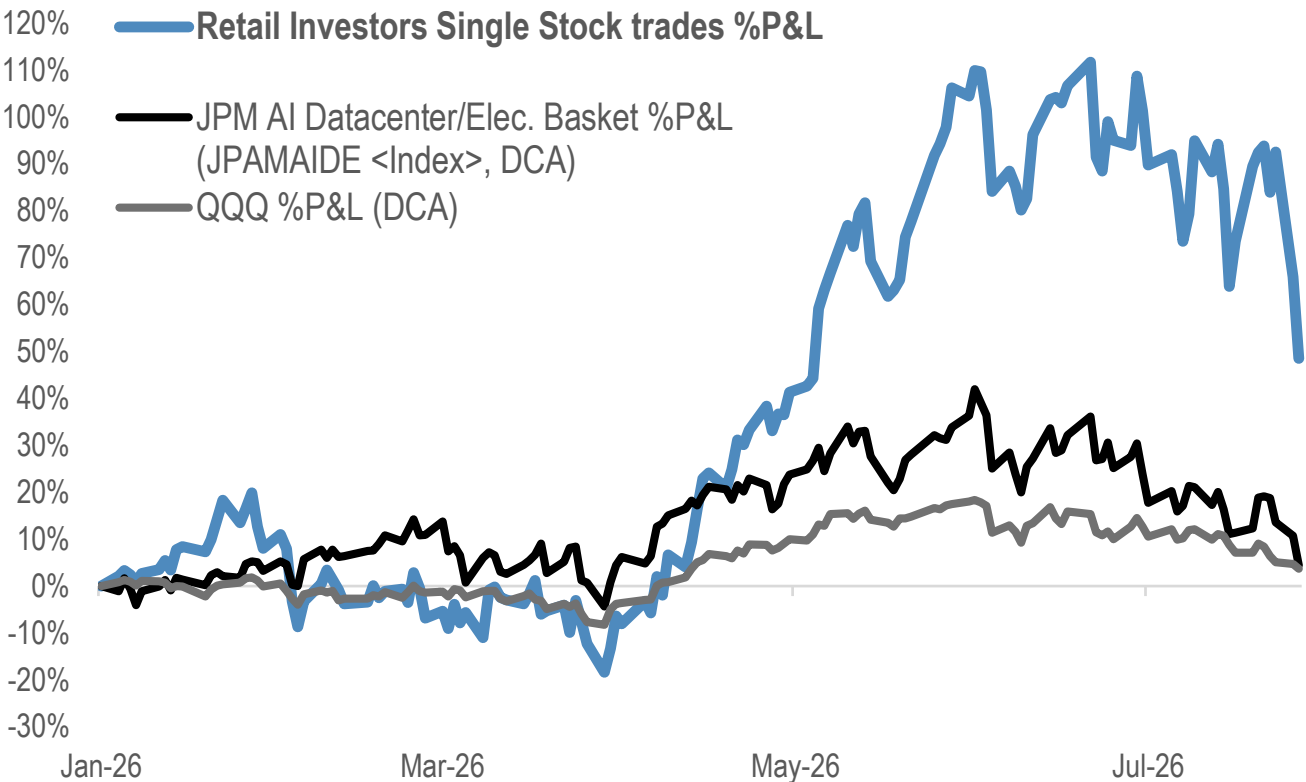
Source: J.P. Morgan Equity Strategy & Quantitative Research

Figure 8: Memory Stocks: Cumulative Retail Net Bought
In \$B



Source: J.P. Morgan Equity Strategy & Quantitative Research

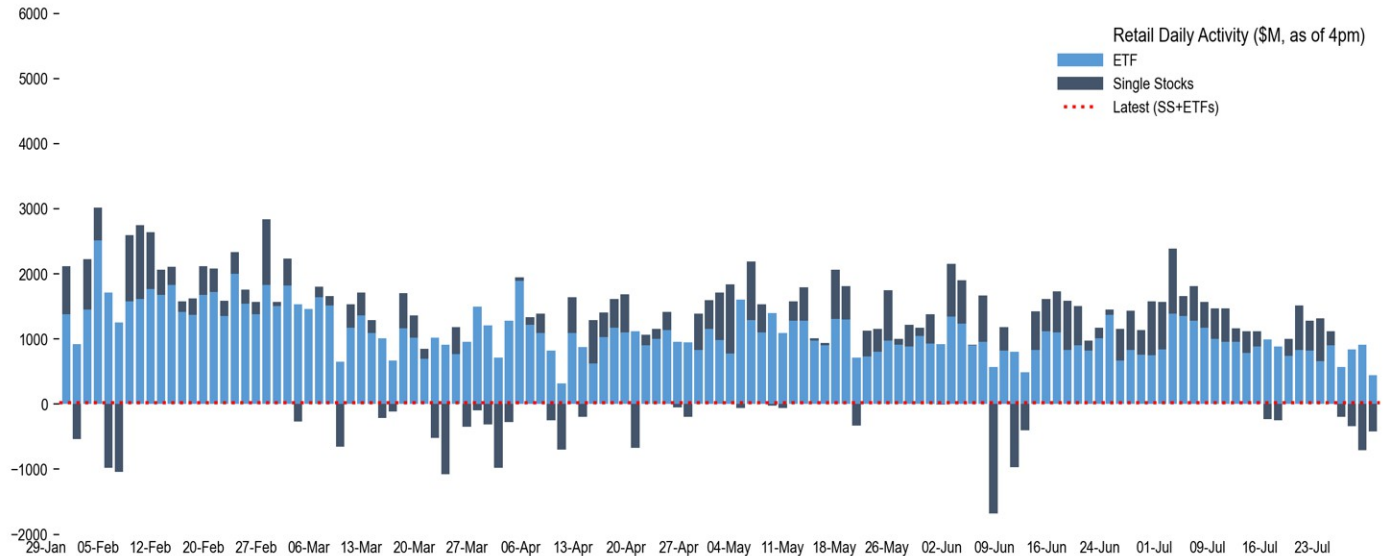
Figure 9: Retail Investors Single Stocks P&L



Source: J.P. Morgan Equity Strategy & Quantitative Research

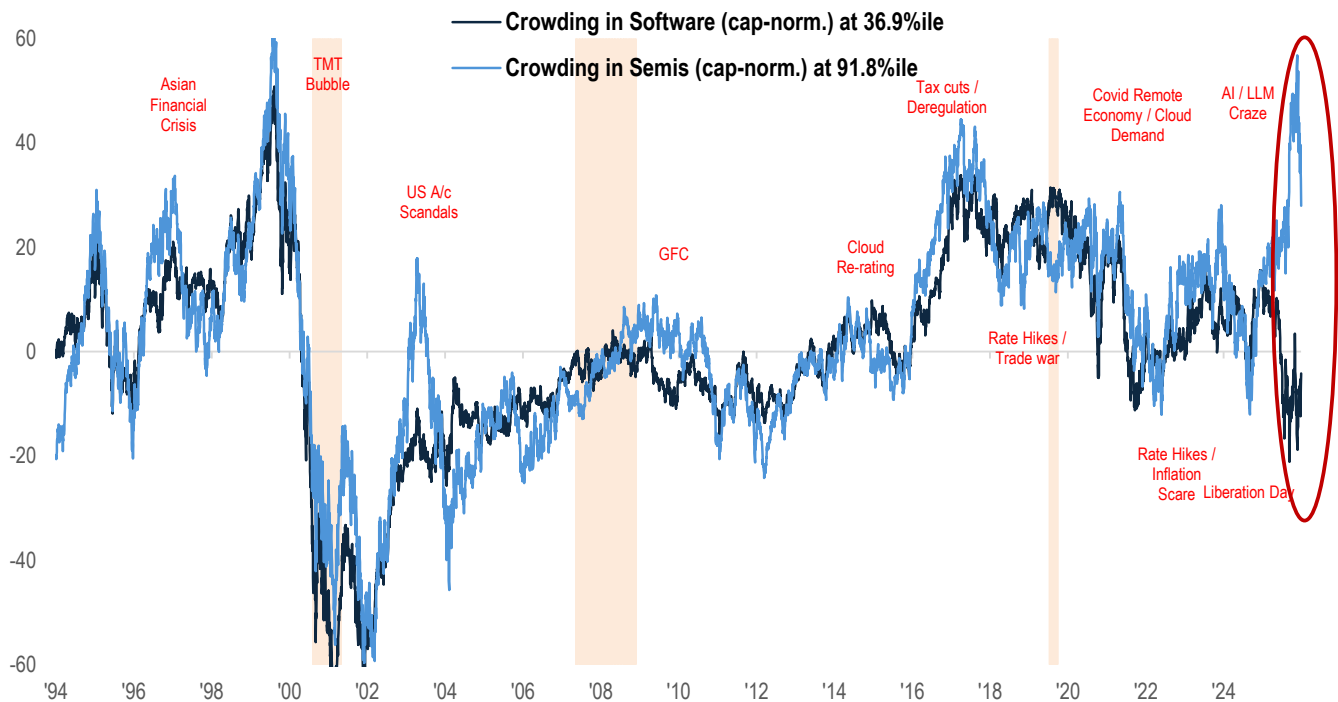
Figure 10: Retail Investor Daily Purchases by Stocks and ETFs

\$M, as of Jul 29th



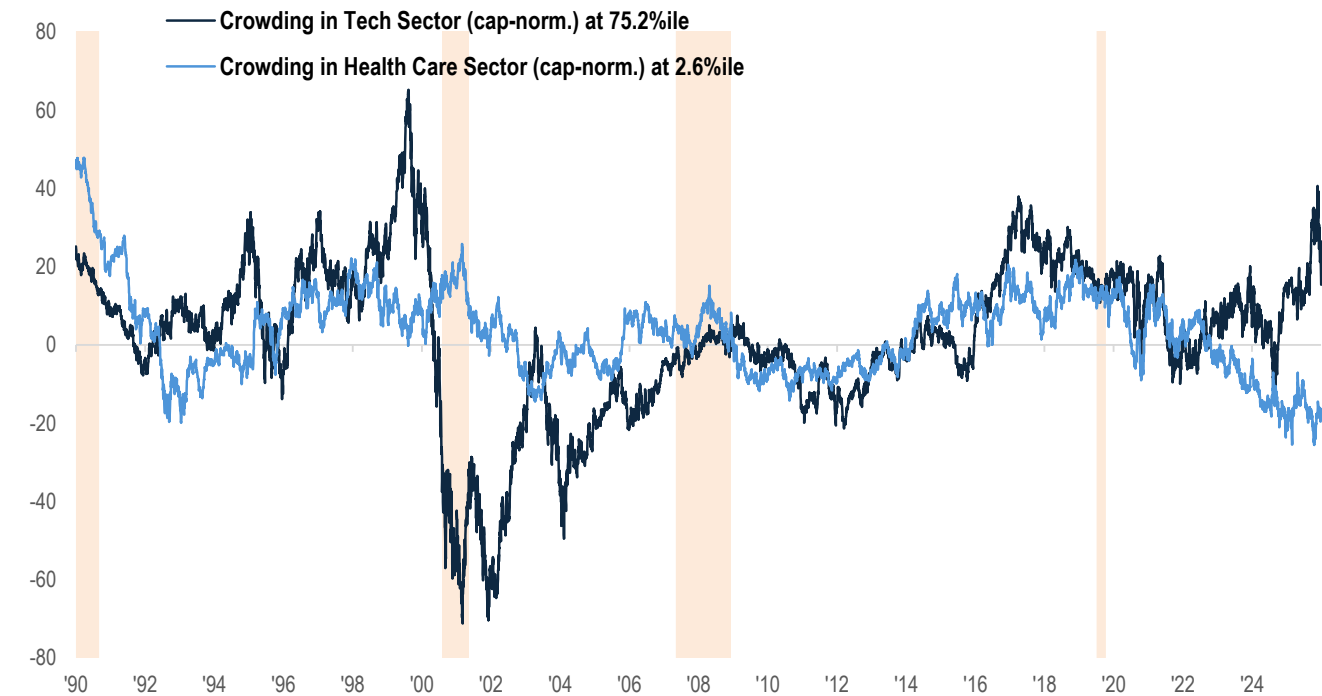
Source: J.P. Morgan Equity Strategy & Quantitative Research

Figure 11: Semis vs Software Crowding



Source: J.P. Morgan Equity Strategy & Quantitative Research

Figure 12: Tech vs Health Care Crowding



Source: J.P. Morgan Equity Strategy & Quantitative Research

Retail Cash Activity – Overview

Figure 13: Retail Single Stock Activity by Themes

Thematic Baskets		Δ in Retail Turnover (\$Bn)		
Basket	Index	Week Ago	Month Ago	YTD
AI/Datacenter/Electrification	JPMAIDE	2.08	5.70	45.48
Magnificent 7	Mag7	2.07	4.30	36.29
S&P 500 Top 30 AI/Datacenter Beneficiaries	JPRAD30	1.79	5.06	44.17
AI Software/Product/Monetization	JPAMISO	1.69	3.57	28.12
Growth	JPAMGROW	1.64	4.48	42.93
OBDBA Immediate Expensing Beneficiaries	JPGTIEXB	1.27	3.02	21.63
Sensitive to Low-End Consumer	JPRLOWCO	-0.04	-0.12	-1.51
US Federal Budget Sensitive	JPAMGOVT	-0.07	-0.11	0.16
US Inflation	JPAMINFL	-0.10	-0.17	-2.08
Inflation Outperformers	JPAMINOP	-0.26	-0.45	-3.70
Trump Deregulation Agenda	JPRGDREG	-0.60	-0.78	-4.80
Wage Sensitive	JPAMWAGG	-0.89	-0.22	3.55

Source: J.P. Morgan Equity Strategy & Quantitative Research. See [report](#), [report](#), and [report](#) for details on above baskets. Note: J.P. Morgan Research does not provide research coverage of the baskets mentioned here, and investors should not expect continuous analysis or additional reports relating to them.

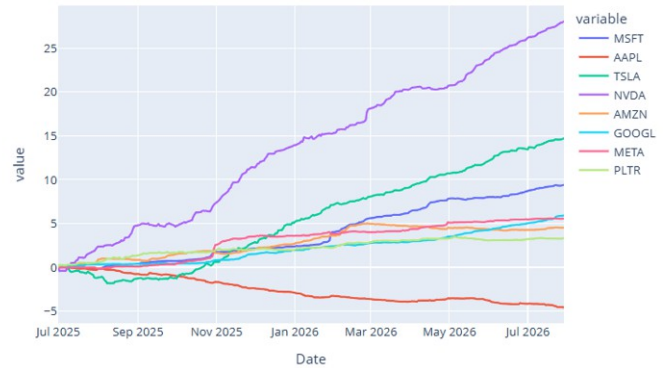
Figure 15: Retail Single Stock Activity by Sector (\$B)

Activity is dominated by Tech and Cons. Disc.



Source: J.P. Morgan Equity Strategy & Quantitative Research

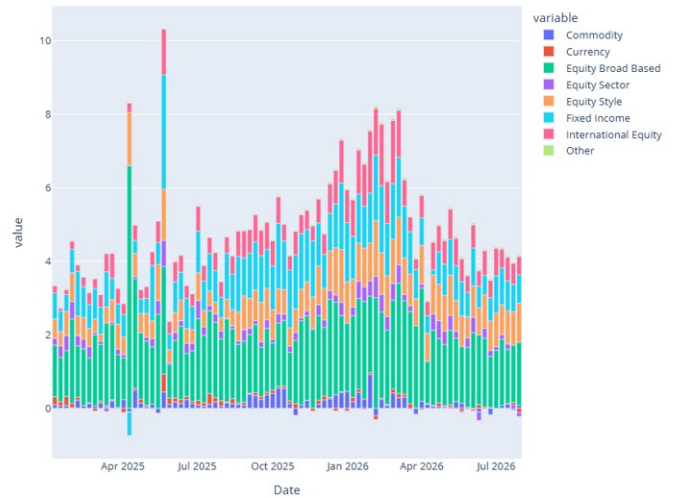
Figure 14: Retail Cumulative Purchases in Mag 7 + PLTR (\$B)



Source: J.P. Morgan Equity Strategy & Quantitative Research

Figure 16: Retail ETF Activity by Themes

Buying / Selling of ETFs aggregated by themes, in \$B.



Source: J.P. Morgan Equity Strategy & Quantitative Research

Single Stock Stories with Elevated Retail Activity

Hedge Funds vs. Retail: High Short Interest Meme Stocks

- **DBGI (\$0.4M bought last week, 0.3z, Figure 19):** The company expanded its secured U.S. program by 32% to \$165M, driven by the addition of new apparel and footwear categories. Retail interest has been strong, with net purchases totaling \$70M since May 2026, amid moderate short interest of ~14% of float, up 10% over the same period.
- **INFQ (\$0.2M sold last week, -0.7z, Figure 20):** The company has undergone a series of management changes and secured three Genesis Mission projects from the U.S. Department of Energy focused on AI-optimized quantum circuit design for nuclear applications. The stock has been on retail investors' radar, with net purchases totaling \$120M since March 2026, alongside elevated short interest of ~16% of float.
- **JOBY (\$13.4M bought last week, -0.0z, Figure 21):** The company signed a binding agreement with Virgin Atlantic, naming it Joby's exclusive airline partner for air taxi services in the UK. Retail investors have purchased \$400M year-to-date, against a backdrop of high short interest of ~18% of float, which has increased by 8% over the same period.

Earnings Stories

- **AAPL (\$18.6M sold on 7/29, -0.0z, Figure 22):** Apple reports earnings tomorrow AMC, with price elasticity of demand the biggest watchpoint for the company across its key hardware product lines following the announcement of a ~20% price increase on average across iPads and Macs, which also raised expectations for the magnitude of price increases likely to be enacted in conjunction with the launch of iPhone 18. However, even as short-term consumer responses to the unprecedented price increases, precipitated by the rise in memory costs, are being debated by investors, our analysts believe there are several drivers to lead the revenue and earnings outcomes to be much more favorable than currently feared by investors ([link](#)).
- **AMZN (\$9.8M bought on 7/29, -0.1z, Figure 24):** Amazon reports tomorrow AMC, with our analysts expecting AWS growth acceleration of low-mid 30%'s, higher capex guidance and retail margin expansion ([link](#)).
- **MSFT (\$23.4M bought on 7/29, -0.2z, Figure 26):** Microsoft's Cloud unit posted a larger-than-expected increase in sales at +43% y/y, with shares rising ~3% in post-market trading. Capex increased +70% y/y to \$41B during the quarter.
- **META (\$2.2M sold on 7/29, -0.4z, Figure 28):** Meta narrowed its 2026 capex guidance to \$130-145B, lifting the low-end from \$125B. Revenue for the quarter came in line with expectations, but next quarter's outlook was lackluster. Shares fell as much as ~10% in post-market trading after the results were published.
- **QCOM (\$6.8M sold on 7/29, -0.6z, Figure 30):** Qualcomm's earnings missed expectations, but top line was better than expected. Additionally, the company expects non-handset revenue growth to accelerate to 60% in fiscal year 2027, as it tries to reduce its dependence on phones with a new push into the data center market.
- **V (\$2.9M sold on 7/29, -0.2z, Figure 32):** Visa reported another impressive quarter highlighted by robust 12% organic growth on healthy drivers including U.S. volume accelerating 2ppts to 10% growth (the fastest rate since late 2018, excluding COVID) and cross border accelerating to 13% growth. While this momentum was partially supported by one-offs like the World Cup, tax refunds and fuel prices, headline results also overcame headwinds like tough currency volatility and incentive comps ([link](#)).
- **NUE (\$0.8M sold on 7/28, -0.4z, Figure 34):** Nucor delivered a strong quarter, beating expectations on earnings and profitability, led by outperformance in its Steel Mills and Raw Materials segments. Results benefited from higher shipments, better margins, and a one-time tariff refund, with utilization running above peers and backlog continuing to build. Downstream operations also contributed through stronger volumes and steady pricing, while construction-related demand drivers (including infrastructure and data centers) remained a key support. Raw Materials improved on higher shipments and better transfer pricing dynamics. The company generated solid free cash flow, used it to fund aggressive buybacks, and the stock rose ~7% on the constructive outlook ([link](#)).
- **INTC (\$5.2M sold on 7/24, -0.3z, Figure 36):** Intel's 2Q results and 3Q guidance cleared expectations by wide margins, reflecting supply unlock (yields, cycle times, wafer starts), record DCAI/server momentum, and firm pricing and mix, but a step-function capex cycle that puts a potential equity/debt raise on the table, memory-driven erosion of the client TAM beginning in 2H26, and an external foundry business still lacking an anchor customer keep our analysts UW even as they materially raise estimates ([link](#)). The stock was down ~8% post-earnings.
- **RTX (\$3.5M sold on 7/24, -0.5z, Figure 38):** Investors seemed pleased with RTX's strong operating performance in 2Q as

the stock outperformed the S&P 8.5% by on earnings day. Defense and aftermarket were especially strong and while Pratt mix debates will likely remain unsettled. Longer term margin expectations for 19-20% at Collins are unchanged and given the growth and incrementals required, this seems achievable at decade end. And of course, among the defense “primes”, Raytheon performance remains quite strong in both absolute and relative terms ([link](#)).

- **DLR (\$12.3M sold on 7/24, -6.0z, Figure 40):** Digital Realty Trust was up +11% after reporting strong 2Q results and raising 2026 guidance for total revenue, adj. EBITDA and core FFO, primarily driven by a robust pricing environment. Subsequent to quarter end, the company signed two leases with hyperscale tenants for \$205M of annualized GAAP rent (\$410M at 100% share), which was excluded from the reported 2Q26 backlog ([link](#)).
- **TMUS (\$2.1M sold on 7/23, -0.3z, Figure 42):** T-Mobile pulled back ~11% on 2Q results, which likely reflected positioning into the print, which anticipated a beat-and-raise. While our analysts trimmed their 2026 service revenue and EBITDA estimates marginally on forward commentary from the call, their long-term estimates remain largely unchanged, supporting an ~8% core EBITDA and ~11% FCF/share CAGR through 2028E, above peers ([link](#)).
- **CSX (\$1.6M bought on 7/22, 1.1z, Figure 44):** CSX proved why sentiment was skewed positive heading into the print with another operating beat and raise, despite service metrics which continued to slip throughout the quarter ([link](#)). The stock was up +6% post-earnings.
- **ROL (\$1.7M sold on 7/22, -1.8z, Figure 46):** Rollins posted a soft 2Q, with the company missing Street expectations across segments. The shortfall versus management’s plan was most pronounced in Residential brands tied to consumer-initiated demand (search, digital media, inbound calls) as lead volumes softened. Commercial and Termite/Ancillary still showed momentum but came in below expectations. Compounding the demand softness, costs were positioned for a stronger peak season that failed to materialize, driving roughly 120 bps of YoY margin contraction ([link](#)). The stock fell ~9%.

Macro & Fundamental Stories

- **SPCX (\$81.2M bought on 7/24, 1.0z, Figure 48):** SpaceX undertook a successful 13th Starship test launch. This was also the second V3 test launch. The rocket achieved its objectives, including deploying 20 Starlink satellites. Importantly, the upper stage landing in the Indian Ocean was relatively soft and controlled, though the booster landing in the Gulf was harder than expected, as not all engines lit for the return to Earth. After the launch, Elon Musk indicated that unless issues emerged from the flight 13 mission data, SpaceX would attempt to catch the second stage in its Mechazilla “chopstick arms” on flight 14. This is a key milestone planned for this year ([link](#)). The stock has continued to underperform, reaching a level ~15%+ below its IPO price.

Meme Battlegrounds: Retail Buying vs. Hedge Fund Shorting

In this section, we are flagging a few stocks that are heavily mentioned on Social Media and have high Retail buying as well as high Hedge fund Shorting. These stocks may experience unexpected flows in case of increased activity.

Social Media Screen and Retail Activity

Figure 17: Most Hyped Stocks on Social Media

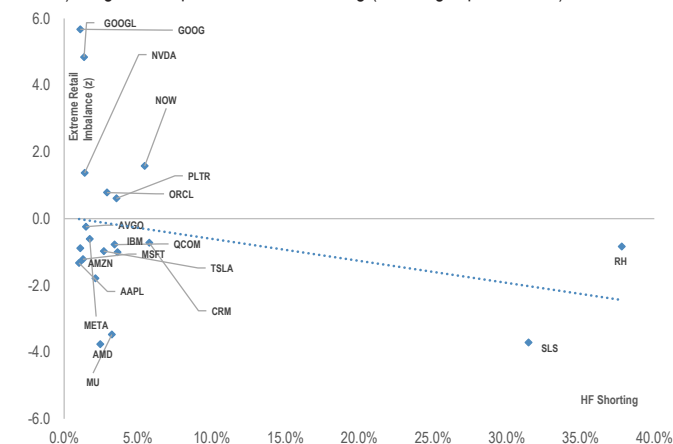
Top 20 names based on social media traction over the week (Social Interest), Scaled by market cap

Ticker	Name	Mkt Cap (\$B)	Social Interest Score	Median Short Interest (% float)	Extreme Imbalance 1D Z	Date of Imbalance
PLTR	Palantir Technologies Inc	296.1	34.0	3.5%	0.6	28-Jul
NVDA	NVIDIA Corp	4767.6	33.0	1.4%	1.4	27-Jul
SLS	SELLAS Life Sciences Group Inc	1.9	31.6	31.5%	-3.7	27-Jul
AMZN	Amazon.com Inc	2483.4	31.4	1.1%	-0.9	27-Jul
META	Meta Platforms Inc	1506.3	31.2	1.7%	-0.6	24-Jul
RH	RH	3.4	30.8	37.8%	-0.8	24-Jul
MSFT	Microsoft Corp	2922.0	30.5	1.3%	-1.2	24-Jul
AMD	Advanced Micro Devices Inc	741.3	30.3	2.5%	-3.8	28-Jul
GOOGL	Alphabet Inc	4074.7	29.9	1.4%	4.8	23-Jul
GOOG	Alphabet Inc	4074.7	29.8	1.1%	5.7	23-Jul
TSLA	Tesla Inc	1214.2	29.3	2.7%	-1.0	24-Jul
AVGO	Broadcom Inc	1812.2	28.2	1.5%	-0.2	28-Jul
AAPL	Apple Inc	4994.9	28.0	1.0%	-1.3	24-Jul
IBM	International Business Machines Corp	214.4	27.0	3.6%	-1.0	27-Jul
MU	Micron Technology Inc	926.7	26.1	3.2%	-3.5	28-Jul
NOW	ServiceNow Inc	114.4	25.3	5.5%	1.6	23-Jul
CRM	Salesforce Inc	148.6	24.9	5.8%	-0.7	23-Jul
QCOM	QUALCOMM Inc	171.7	24.8	3.4%	-0.8	28-Jul
ORCL	Oracle Corp	345.5	24.1	2.9%	0.8	28-Jul
AMAT	Applied Materials Inc	378.3	24.0	2.1%	-1.8	28-Jul

Source: J.P. Morgan Equity Strategy & Quantitative Research, S3 Partners

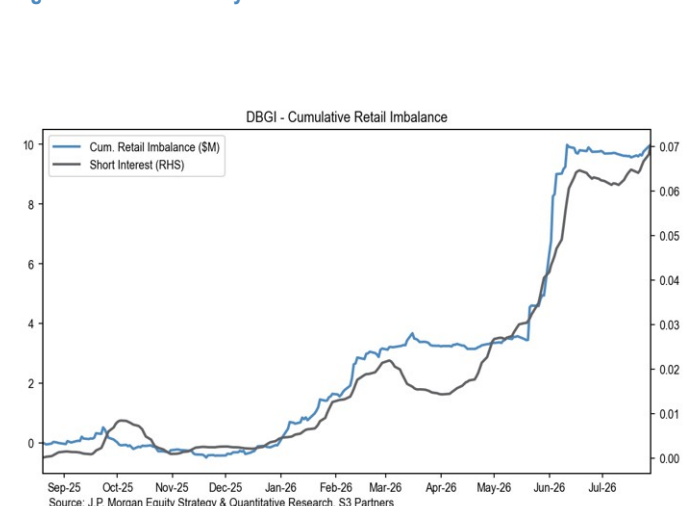
Figure 18: Decreasing Squeeze Risk in Meme Stocks: Retail Selling in High Short Interest Names

Positive Slope = Opposite Positioning (Heightened Risk of Short Squeeze or Retail Losses); Negative Slope = Similar Positioning (Diffusing Squeeze Risk)



Source: J.P. Morgan Equity Strategy & Quantitative Research, S3 Partners

Figure 19: Retail Activity in DBGI



Source: J.P. Morgan Equity Strategy & Quantitative Research, S3 Partners

Figure 20: Retail Activity in INFQ

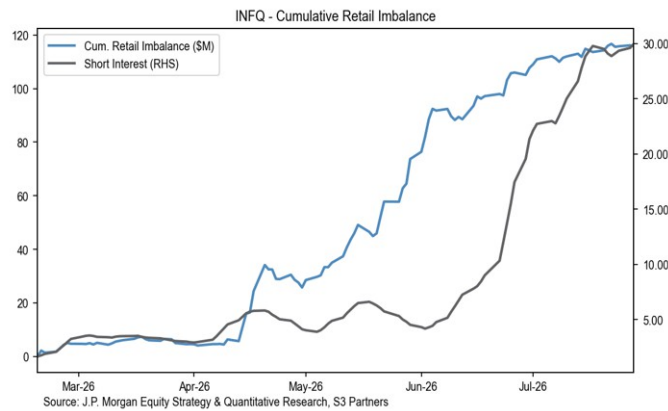
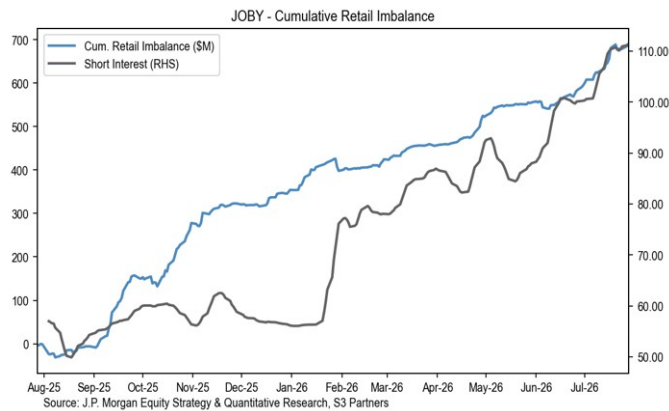


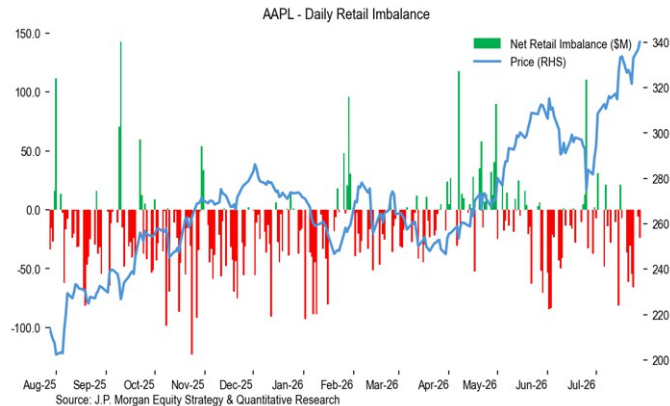
Figure 21: Retail Activity in JOBY



Macro & Fundamental Stories

Figure 22: Retail Imbalance in AAPL

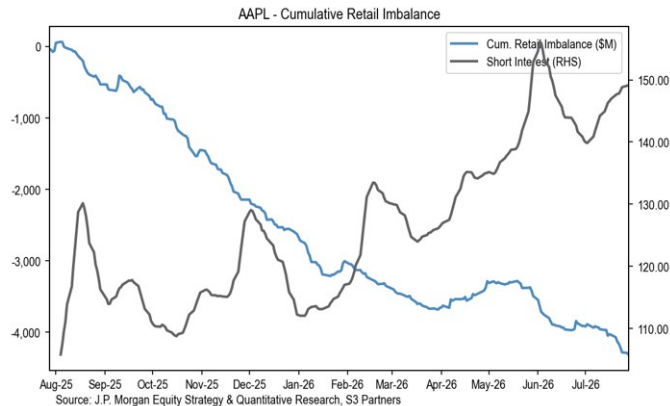
As of July 28th



Source: J.P. Morgan Equity Strategy & Quantitative Research, S3 Partners

Figure 23: Cumulative Retail Imbalance in AAPL

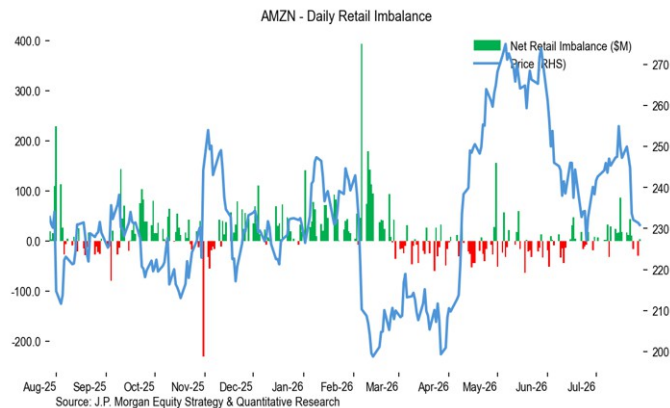
As of July 28th



Source: J.P. Morgan Equity Strategy & Quantitative Research, S3 Partners

Figure 24: Retail Imbalance in AMZN

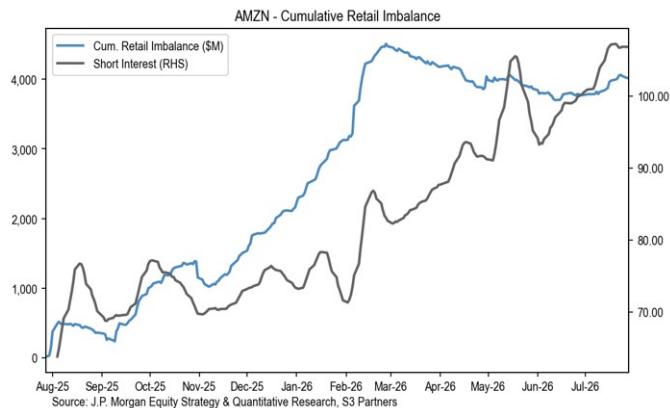
As of July 28th



Source: J.P. Morgan Equity Strategy & Quantitative Research, S3 Partners

Figure 25: Cumulative Retail Imbalance in AMZN

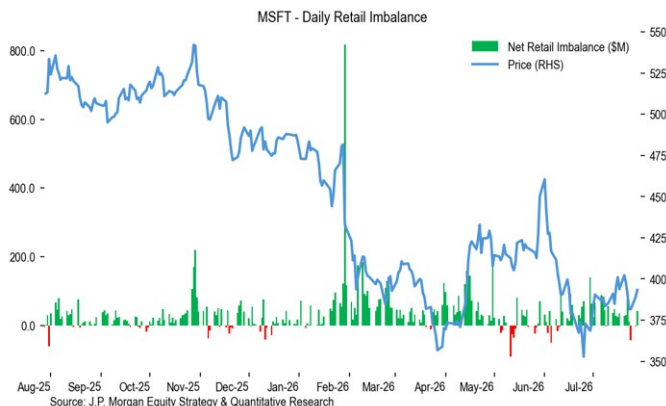
As of July 28th



Source: J.P. Morgan Equity Strategy & Quantitative Research, S3 Partners

Figure 26: Retail Imbalance in MSFT

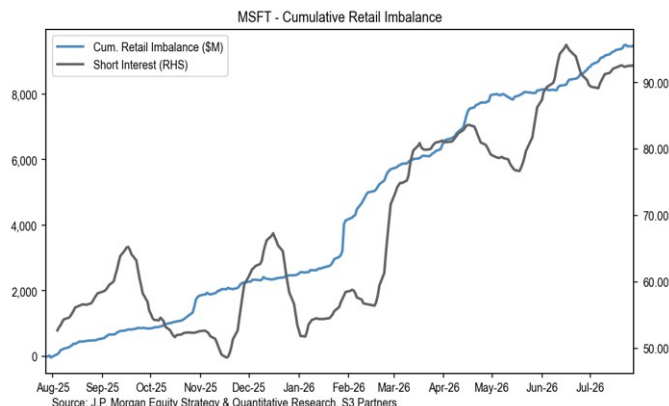
As of July 28th



Source: J.P. Morgan Equity Strategy & Quantitative Research, S3 Partners

Figure 27: Cumulative Retail Imbalance in MSFT

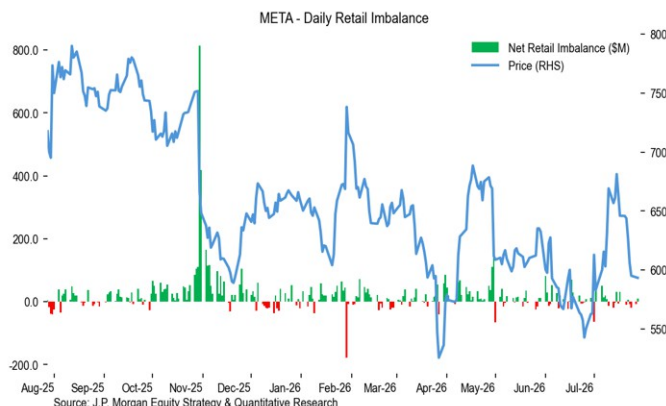
As of July 28th



Source: J.P. Morgan Equity Strategy & Quantitative Research, S3 Partners

Figure 28: Retail Imbalance in META

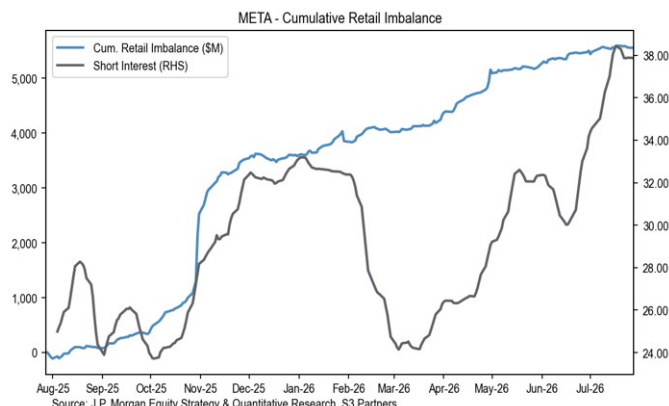
As of July 28th



Source: J.P. Morgan Equity Strategy & Quantitative Research, S3 Partners

Figure 29: Cumulative Retail Imbalance in META

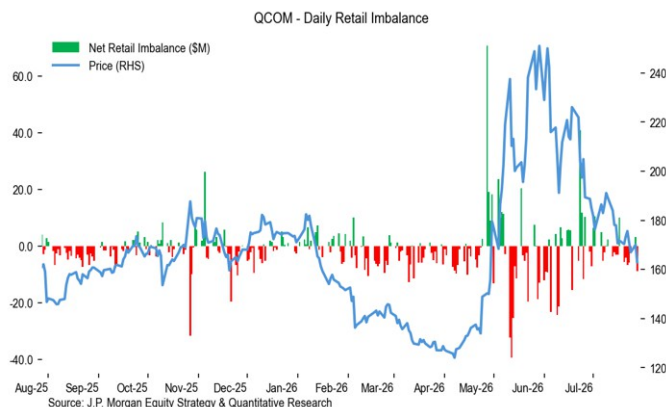
As of July 28th



Source: J.P. Morgan Equity Strategy & Quantitative Research, S3 Partners

Figure 30: Retail Imbalance in QCOM

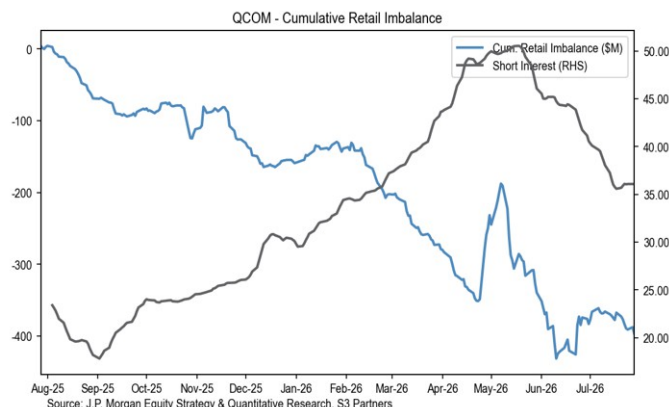
As of July 28th



Source: J.P. Morgan Equity Strategy & Quantitative Research, S3 Partners

Figure 31: Cumulative Retail Imbalance in QCOM

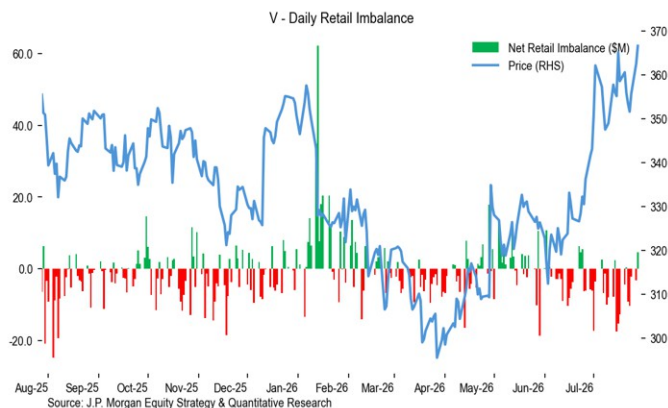
As of July 28th



Source: J.P. Morgan Equity Strategy & Quantitative Research, S3 Partners

Figure 32: Retail Imbalance in V

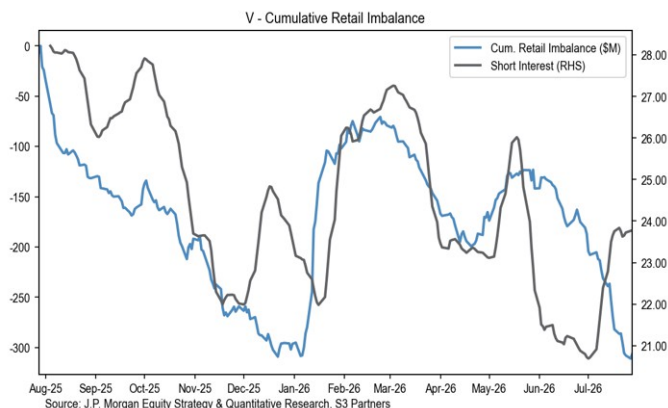
As of July 28th



Source: J.P. Morgan Equity Strategy & Quantitative Research, S3 Partners

Figure 33: Cumulative Retail Imbalance in V

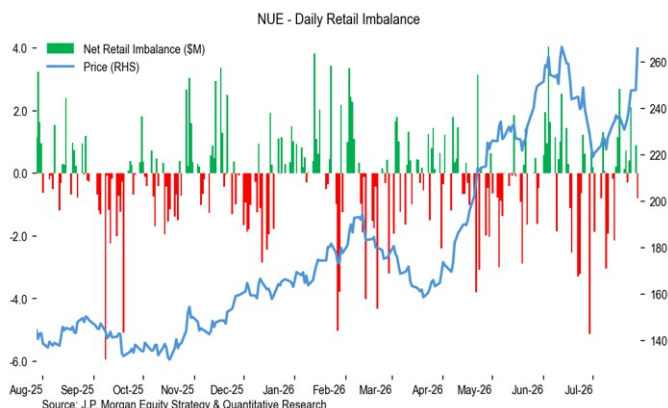
As of July 28th



Source: J.P. Morgan Equity Strategy & Quantitative Research, S3 Partners

Figure 34: Retail Imbalance in NUE

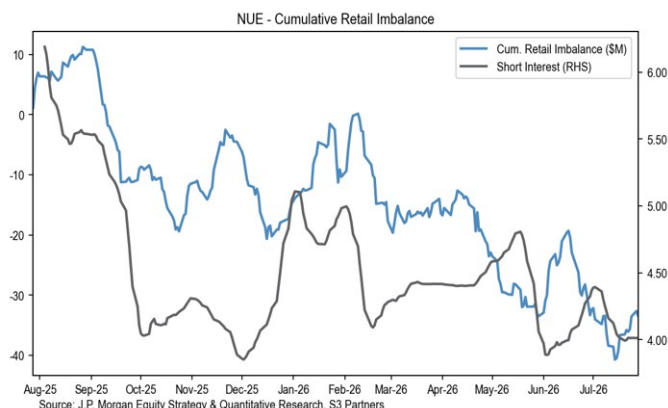
As of July 28th



Source: J.P. Morgan Equity Strategy & Quantitative Research, S3 Partners

Figure 35: Cumulative Retail Imbalance in NUE

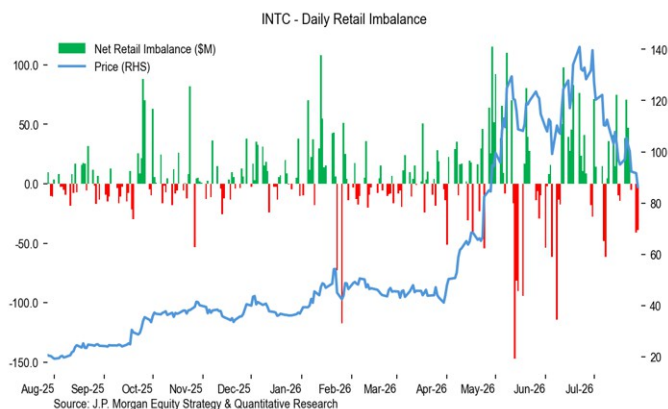
As of July 28th



Source: J.P. Morgan Equity Strategy & Quantitative Research, S3 Partners

Figure 36: Retail Imbalance in INTC

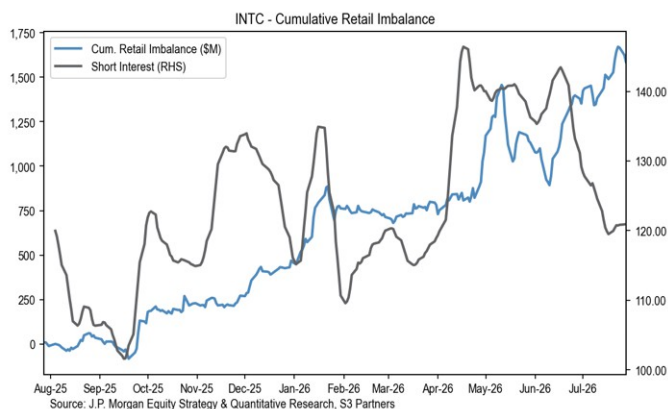
As of July 28th



Source: J.P. Morgan Equity Strategy & Quantitative Research, S3 Partners

Figure 37: Cumulative Retail Imbalance in INTC

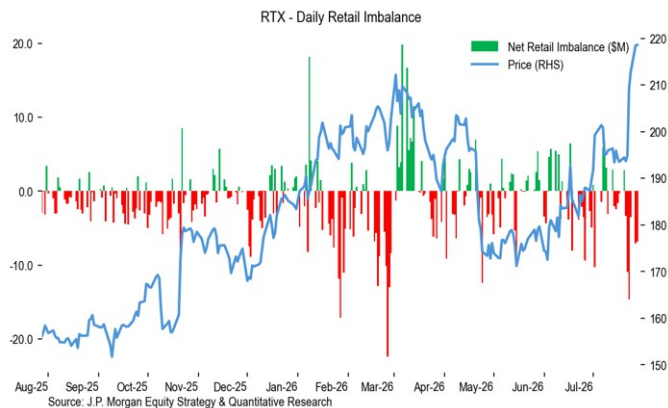
As of July 28th



Source: J.P. Morgan Equity Strategy & Quantitative Research, S3 Partners

Figure 38: Retail Imbalance in RTX

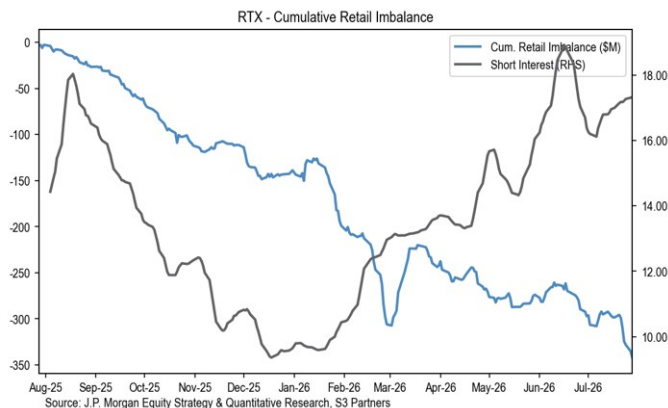
As of July 28th



Source: J.P. Morgan Equity Strategy & Quantitative Research, S3 Partners

Figure 39: Cumulative Retail Imbalance in RTX

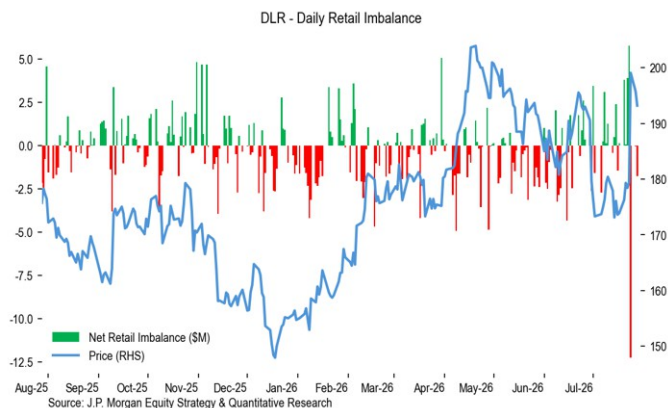
As of July 28th



Source: J.P. Morgan Equity Strategy & Quantitative Research, S3 Partners

Figure 40: Retail Imbalance in DLR

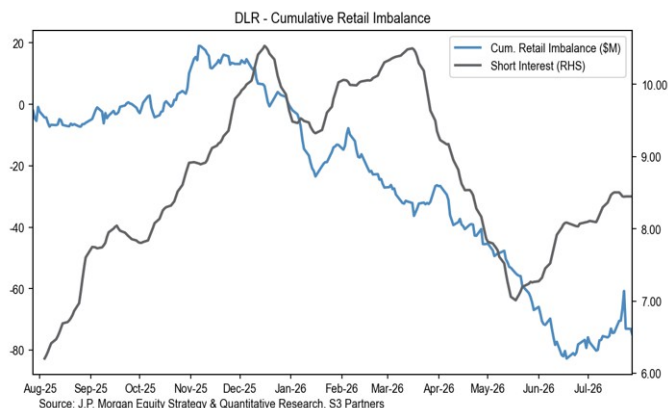
As of July 28th



Source: J.P. Morgan Equity Strategy & Quantitative Research, S3 Partners

Figure 41: Cumulative Retail Imbalance in DLR

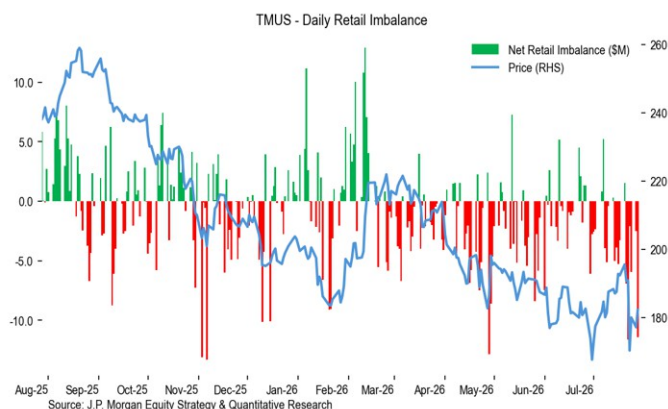
As of July 28th



Source: J.P. Morgan Equity Strategy & Quantitative Research, S3 Partners

Figure 42: Retail Imbalance in TMUS

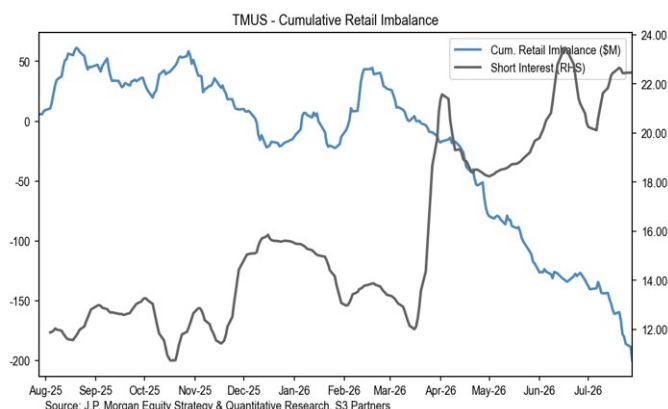
As of July 28th



Source: J.P. Morgan Equity Strategy & Quantitative Research, S3 Partners

Figure 43: Cumulative Retail Imbalance in TMUS

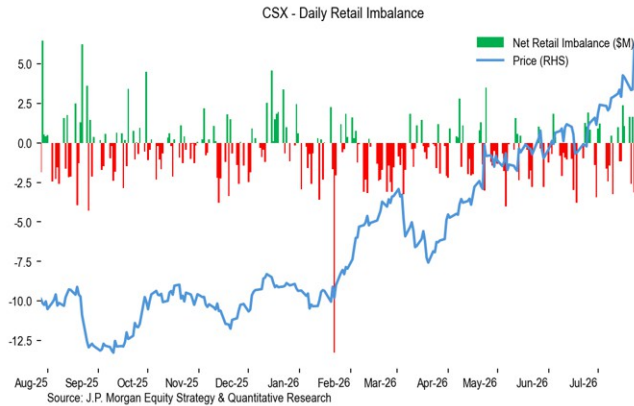
As of July 28th



Source: J.P. Morgan Equity Strategy & Quantitative Research, S3 Partners

Figure 44: Retail Imbalance in CSX

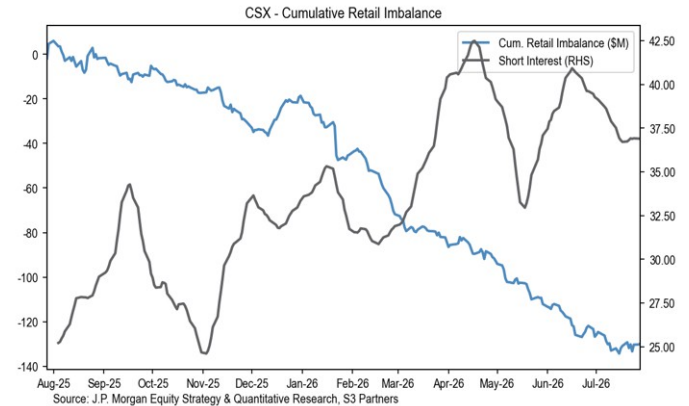
As of July 28th



Source: J.P. Morgan Equity Strategy & Quantitative Research, S3 Partners

Figure 45: Cumulative Retail Imbalance in CSX

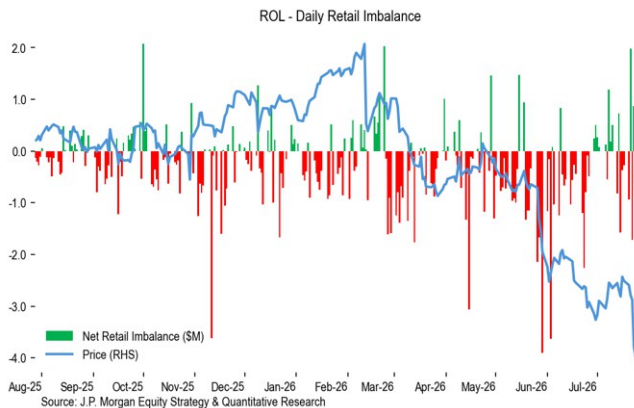
As of July 28th



Source: J.P. Morgan Equity Strategy & Quantitative Research, S3 Partners

Figure 46: Retail Imbalance in ROL

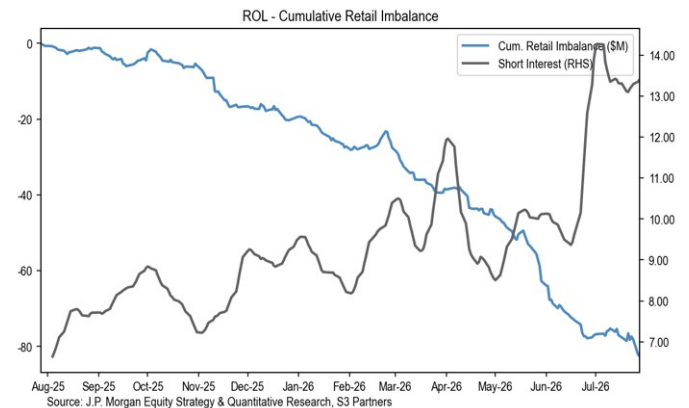
As of July 28th



Source: J.P. Morgan Equity Strategy & Quantitative Research, S3 Partners

Figure 47: Cumulative Retail Imbalance in ROL

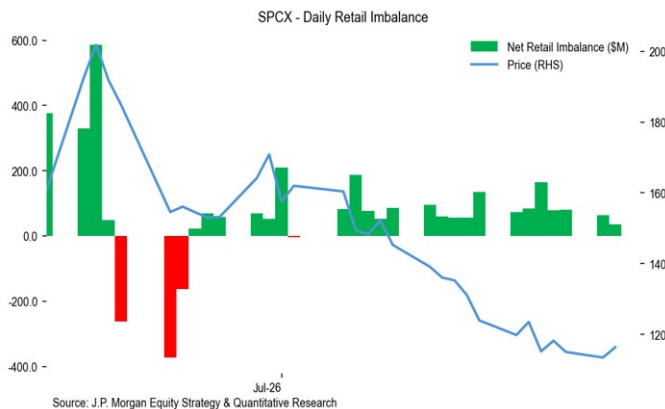
As of July 28th



Source: J.P. Morgan Equity Strategy & Quantitative Research, S3 Partners

Figure 48: Retail Imbalance in SPCX

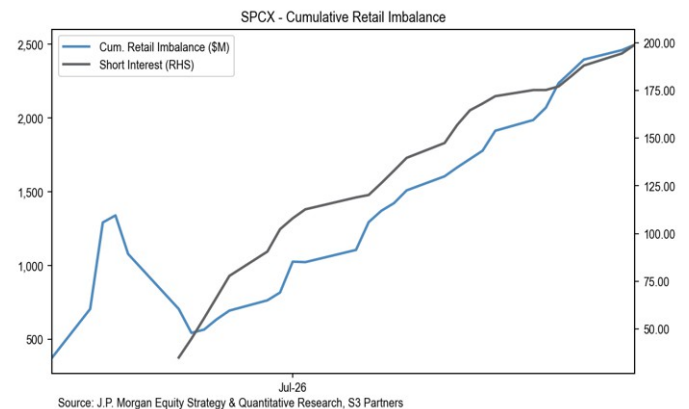
As of July 28th



Source: J.P. Morgan Equity Strategy & Quantitative Research, S3 Partners

Figure 49: Cumulative Retail Imbalance in SPCX

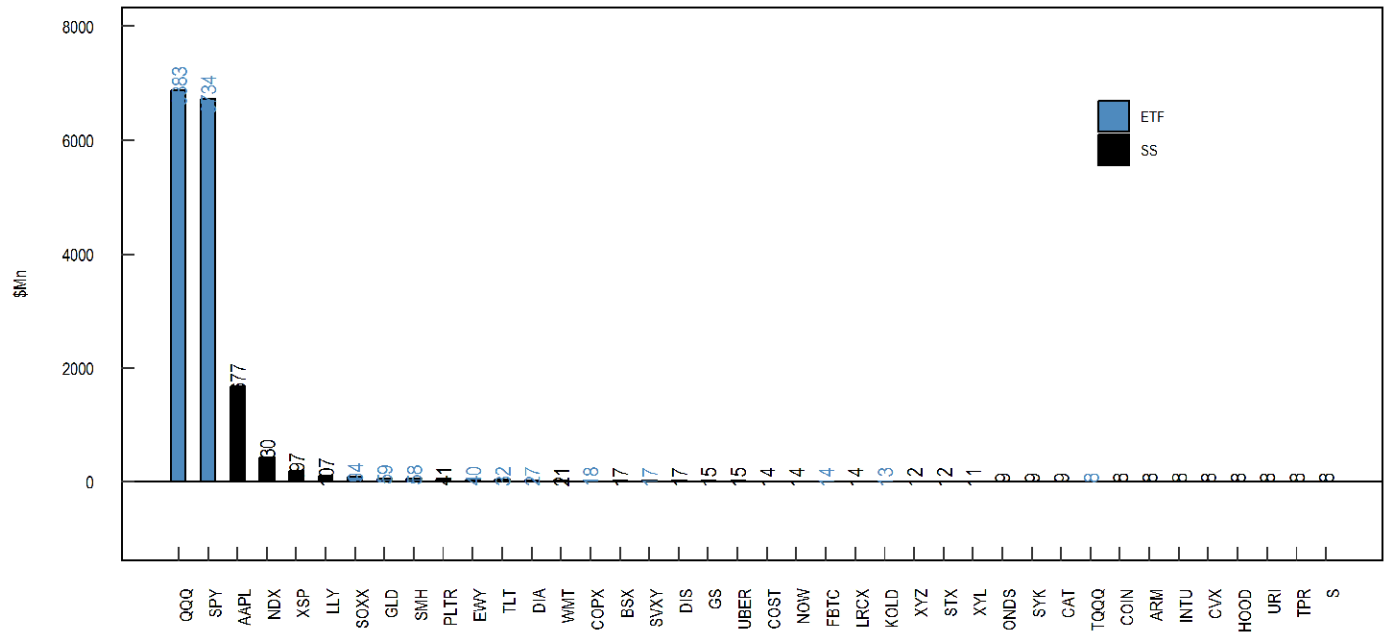
As of July 28th



Source: J.P. Morgan Equity Strategy & Quantitative Research, S3 Partners

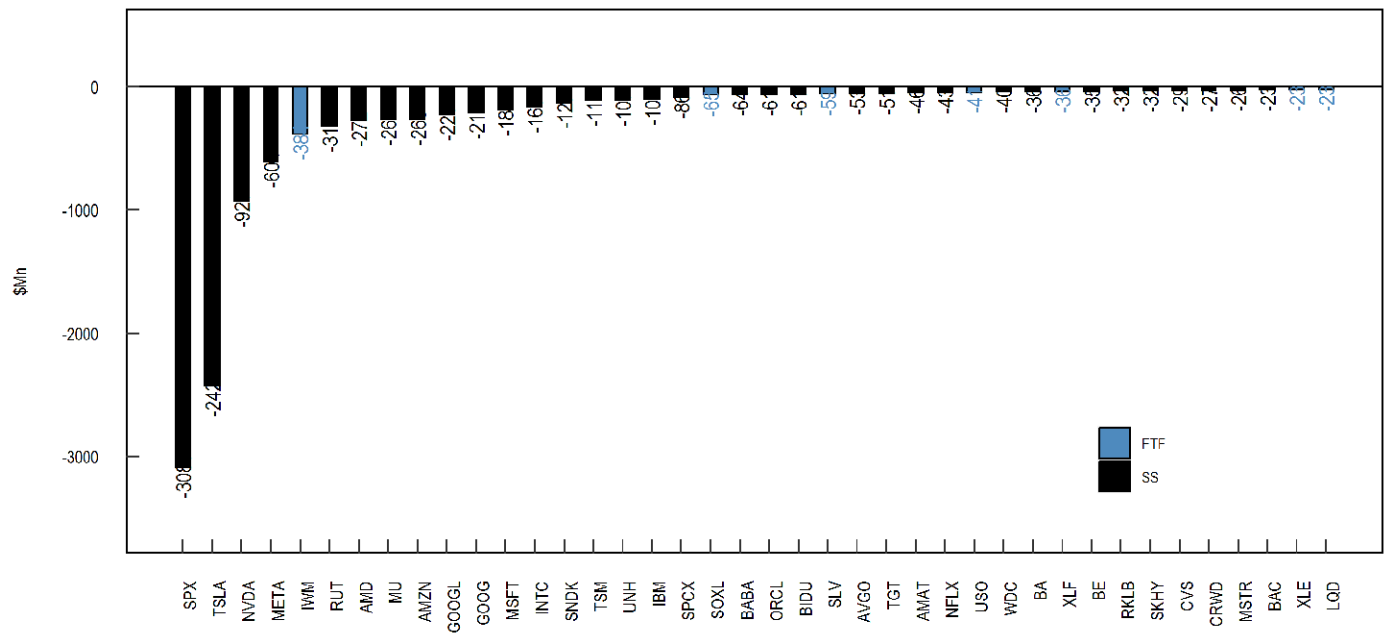
Retail Activity in Options

Figure 50: Top 40 Equities with Most Delta Bought (in \$Mn)



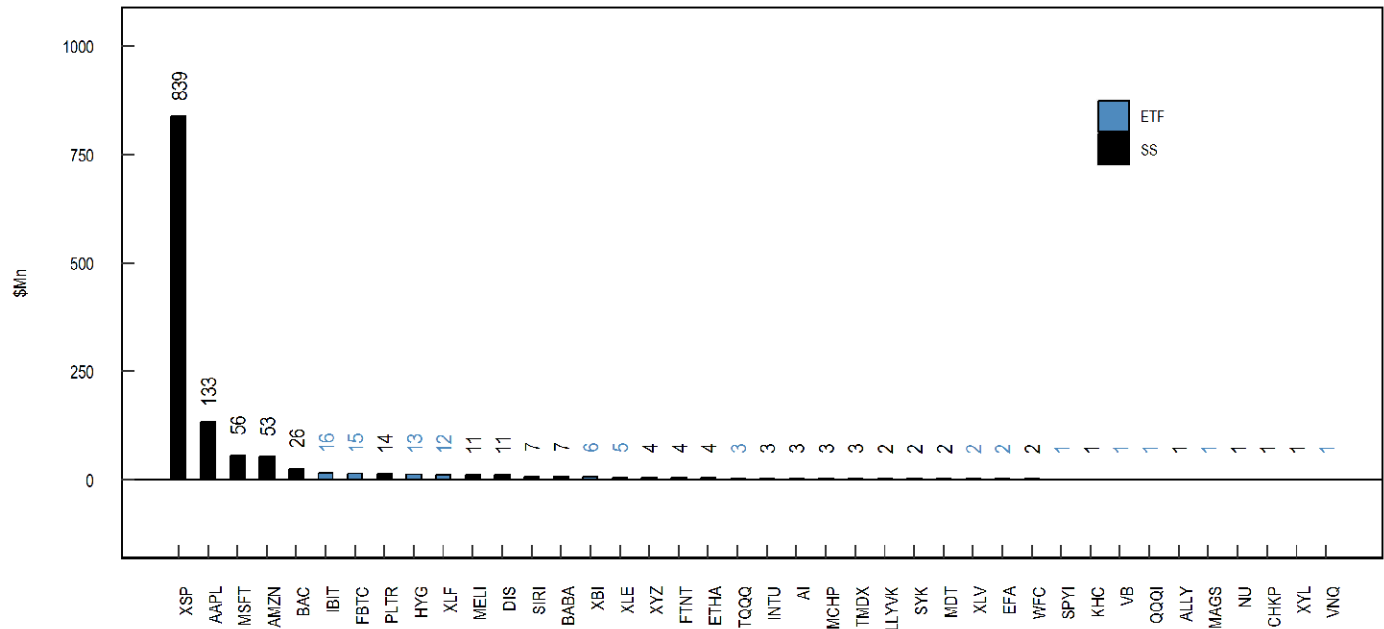
Source: J.P. Morgan Equity Strategy & Quantitative Research

Figure 51: Top 40 Equities with Most Delta Sold (in \$Mn)



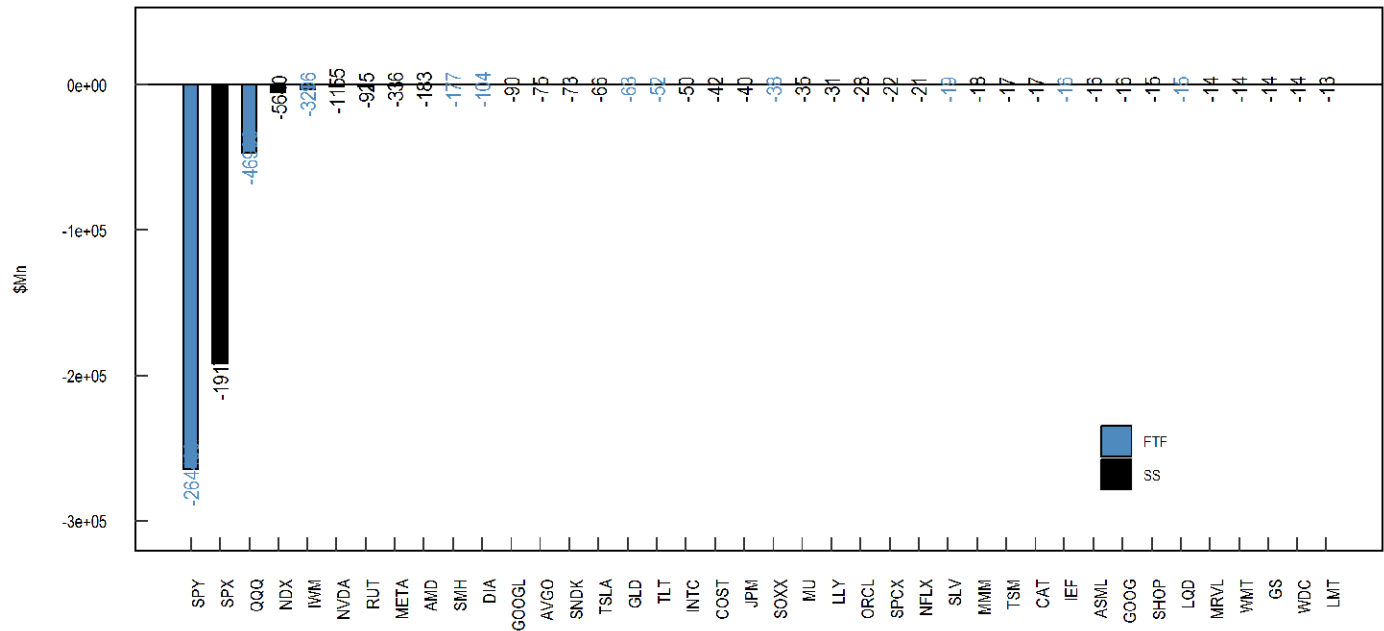
Source: J.P. Morgan Equity Strategy & Quantitative Research

Figure 52: Top 40 Equities with Most Gamma Bought (in \$Mn)



Source: J.P. Morgan Equity Strategy & Quantitative Research

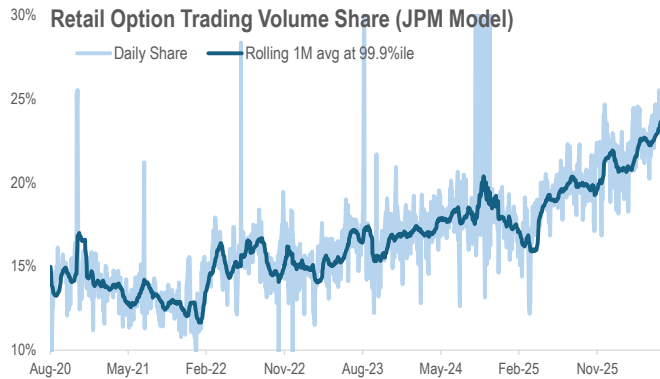
Figure 53: Top 40 Equities with Most Gamma Sold (in \$Mn)



Source: J.P. Morgan Equity Strategy & Quantitative Research

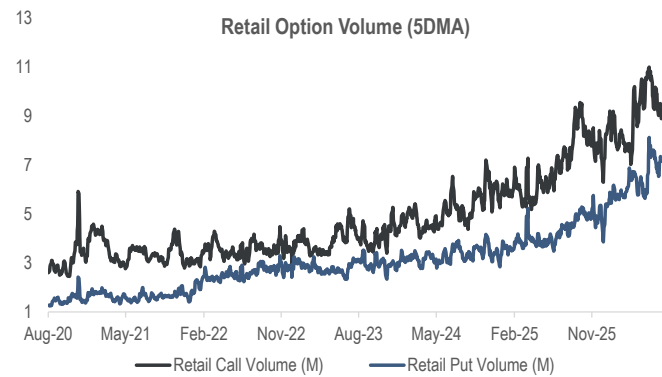
Retail Investors Options Activity by Sectors

Figure 54: Retail Options Trading Share



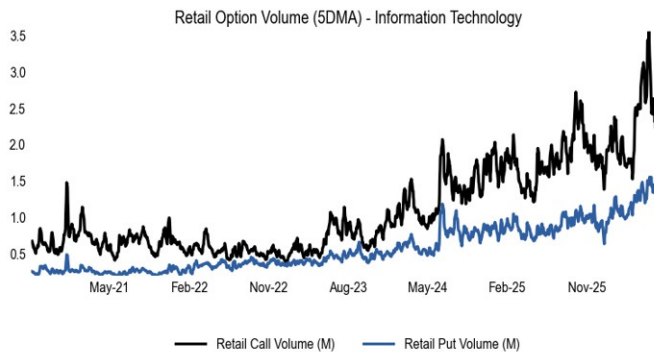
Source: J.P. Morgan Equity Strategy & Quantitative Research

Figure 55: Retail Options Volume (Calls and Puts)



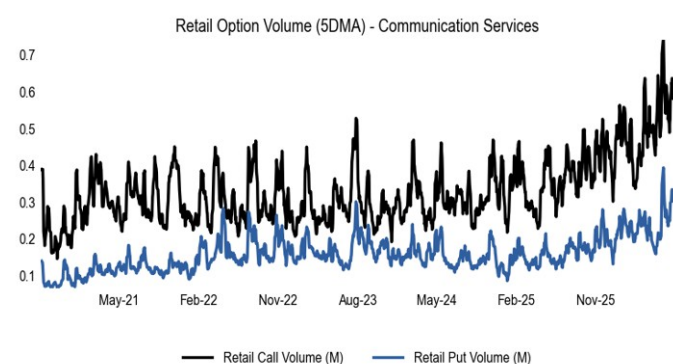
Source: J.P. Morgan Equity Strategy & Quantitative Research

Figure 56: Retail Options Volume (Calls and Puts), Information Technology



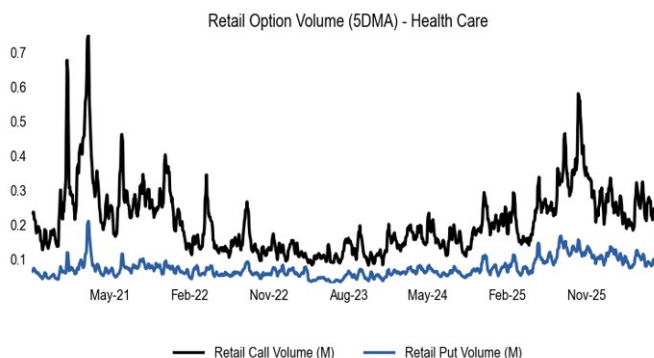
Source: J.P. Morgan Equity Strategy & Quantitative Research

Figure 57: Retail Options Volume (Calls and Puts), Communication Services



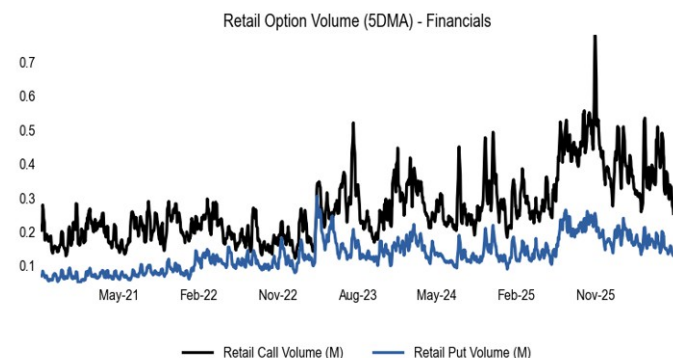
Source: J.P. Morgan Equity Strategy & Quantitative Research

Figure 58: Retail Options Volume (Calls and Puts), Health Care



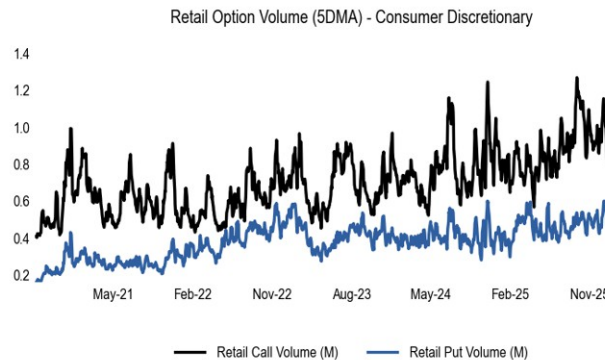
Source: J.P. Morgan Equity Strategy & Quantitative Research

Figure 59: Retail Options Volume (Calls and Puts), Financials



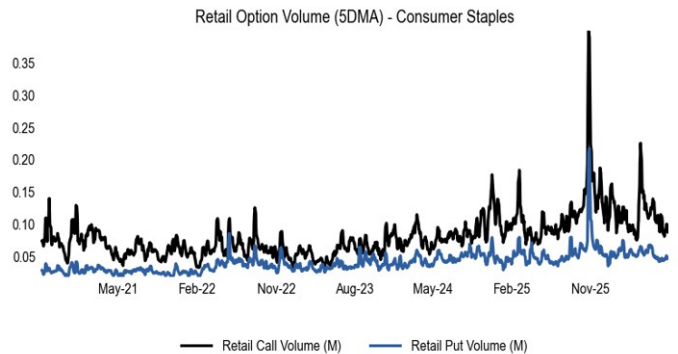
Source: J.P. Morgan Equity Strategy & Quantitative Research

Figure 60: Retail Options Volume (Calls and Puts), Discretionary



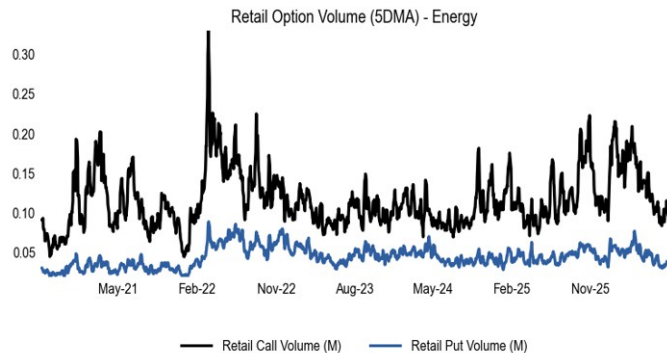
Source: J.P. Morgan Equity Strategy & Quantitative Research

Figure 61: Retail Options Volume (Calls and Puts), Staples



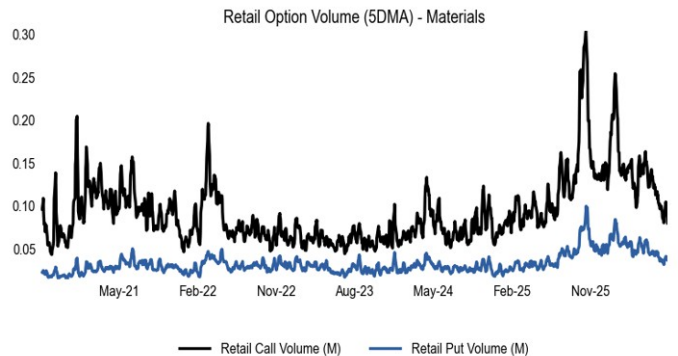
Source: J.P. Morgan Equity Strategy & Quantitative Research

Figure 62: Retail Options Volume (Calls and Puts), Energy



Source: J.P. Morgan Equity Strategy & Quantitative Research

Figure 63: Retail Options Volume (Calls and Puts), Materials



Source: J.P. Morgan Equity Strategy & Quantitative Research

Appendix: Alternative Data Points on Retail Activity

Figure 64: Retail Brokerage Volume Hit Record ~37% in January 2021

Source Flow by Retail Brokerage, % of Overall US Equity Mkt Volume (Stock + ETF)

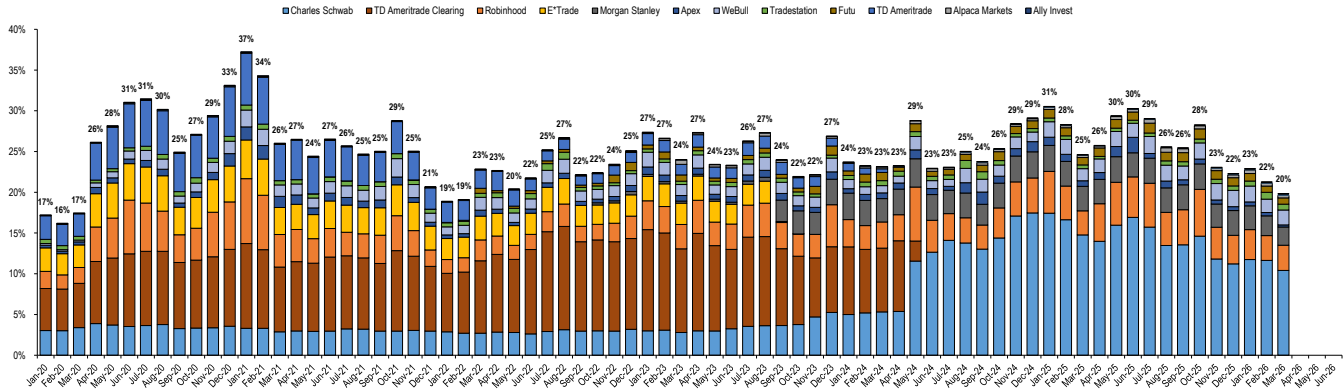
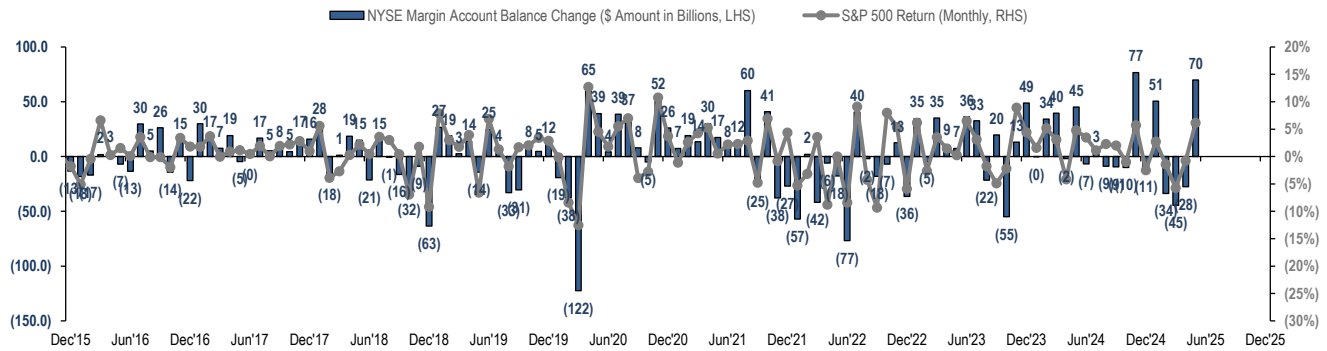


Figure 65: NYSE Margin Account – Monthly Net Debit Balance Change

\$ in Billions, Net Debit balance equals margin debit balance less total credit balances



Source: J.P. Morgan Equity Strategy & Quantitative Research, Bloomberg Finance L.P., FINRA, Quarterly 606 Reports (for two figures above).

Companies Discussed in This Report (all prices in this report as of market close on 29 July 2026, unless otherwise indicated)
Intel(INTC/\$81.88/UW)

Analyst Certification: The Research Analyst(s) denoted by an “AC” on the cover of this report certifies (or, where multiple Research Analysts are primarily responsible for this report, the Research Analyst denoted by an “AC” on the cover or within the document individually certifies, with respect to each security or issuer that the Research Analyst covers in this research) that: (1) all of the views expressed in this report accurately reflect the Research Analyst’s personal views about any and all of the subject securities or issuers; and (2) no part of any of the Research Analyst’s compensation was, is, or will be directly or indirectly related to the specific recommendations or views expressed by the Research Analyst(s) in this report. For all Korea-based Research Analysts listed on the front cover, if applicable, they also certify, as per KOFIA requirements, that the Research Analyst’s analysis was made in good faith and that the views reflect the Research Analyst’s own opinion, without undue influence or intervention.

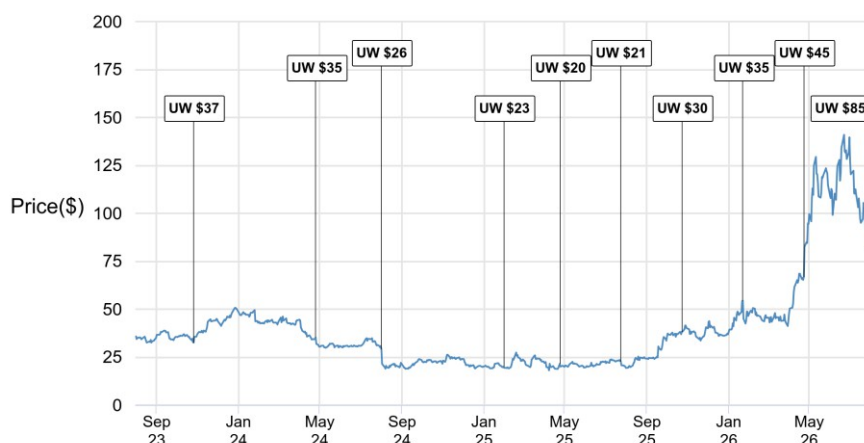
All authors named within this report are Research Analysts who produce independent research unless otherwise specified. In Europe, Sector Specialists (Sales and Trading) may be shown on this report as contacts but are not authors of the report or part of the Research Department.

Important Disclosures

- **Market Maker:** J.P. Morgan Securities LLC makes a market in the securities of Intel or related entities.
- **Market Maker/ Liquidity Provider:** J.P. Morgan is a market maker and/or liquidity provider in the financial instruments of/related to Intel or related entities.
- **Manager or Co-manager:** J.P. Morgan acted as manager or co-manager in a public offering of securities or financial instruments (as such term is defined in Directive 2014/65/EU) of/for Intel or related entities within the past 12 months.
- **Beneficial Ownership (1% or more):** J.P. Morgan beneficially owns 1% or more of a class of common equity securities of Intel or related entities.
- **Client:** J.P. Morgan currently has, or had within the past 12 months, the following entity(ies) as clients: Intel or related entities.
- **Client/Investment Banking:** J.P. Morgan currently has, or had within the past 12 months, the following entity(ies) as investment banking clients: Intel or related entities.
- **Client/Non-Investment Banking, Securities-Related:** J.P. Morgan currently has, or had within the past 12 months, the following entity(ies) as clients, and the services provided were non-investment-banking, securities-related: Intel or related entities.
- **Client/Non-Securities-Related:** J.P. Morgan currently has, or had within the past 12 months, the following entity(ies) as clients, and the services provided were non-securities-related: Intel or related entities.
- **Investment Banking Compensation Received:** J.P. Morgan has received in the past 12 months compensation for investment banking services from Intel or related entities.
- **Potential Investment Banking Compensation:** J.P. Morgan expects to receive, or intends to seek, compensation for investment banking services in the next three months from Intel or related entities.
- **Non-Investment Banking Compensation Received:** J.P. Morgan has received compensation in the past 12 months for products or services other than investment banking from Intel or related entities.
- **Debt Position:** J.P. Morgan may hold a position in the debt securities of Intel or related entities, if any.

Company-Specific Disclosures: J.P. Morgan does and seeks to do business with companies covered in its research reports. As a result, investors should be aware that the firm may have a conflict of interest that could affect the objectivity of this report. Investors should consider this report as only a single factor in making their investment decision. Important disclosures, including price charts and credit opinion history tables (if applicable), are available for compendium reports and all J.P. Morgan-covered companies, and certain non-covered companies, by visiting <https://www.jpmm.com/research/disclosures>, calling 1-800-477-0406, or e-mailing research.disclosure.inquiries@jpmorgan.com with your request.

Intel (INTC, INTC US) Price Chart



Date	Rating	Price (\$)	Price Target (\$)
27-Oct-23	UW	32.52	37
26-Apr-24	UW	35.11	35
02-Aug-24	UW	29.05	26
31-Jan-25	UW	20.01	23
25-Apr-25	UW	21.49	20
25-Jul-25	UW	22.63	21
24-Oct-25	UW	38.16	30
23-Jan-26	UW	54.32	35
24-Apr-26	UW	66.78	45
24-Jul-26	UW	100.23	85

Source: Bloomberg Finance L.P. and J.P. Morgan; price data adjusted for stock splits and dividends.
Initiated coverage Sep 12, 2002. All share prices are as of market close on the previous business day.

The chart(s) show J.P. Morgan's continuing coverage of the stocks; the current analysts may or may not have covered it over the entire period.
J.P. Morgan ratings or designations: OW = Overweight, N= Neutral, UW = Underweight, NR = Not Rated

Explanation of Equity Research Ratings, Designations and Analyst(s) Coverage Universe:

J.P. Morgan uses the following rating system: Overweight (over the duration of the price target indicated in this report, we expect this stock will outperform the average total return of the stocks in the Research Analyst's, or the Research Analyst's team's, coverage universe); Neutral (over the duration of the price target indicated in this report, we expect this stock will perform in line with the average total return of the stocks in the Research Analyst's, or the Research Analyst's team's, coverage universe); and Underweight (over the duration of the price target indicated in this report, we expect this stock will underperform the average total return of the stocks in the Research Analyst's, or the Research Analyst's team's, coverage universe. NR is Not Rated. In this case, J.P. Morgan has removed the rating and, if applicable, the price target, for this stock because of either a lack of a sufficient fundamental basis or for legal, regulatory or policy reasons. The previous rating and, if applicable, the price target, no longer should be relied upon. An NR designation is not a recommendation or a rating. Some stocks under coverage have a rating but no price target; in these cases, we expect the stock will outperform/perform in line/underperform the average total return of the stocks in the Research Analyst's, or the Research Analyst's team's, coverage universe of the relevant duration of the region. In our Asia (ex-Australia and ex-India) and U.K. small- and mid-cap Equity Research, each stock's expected total return is compared to the expected total return of a benchmark country market index, not to those Research Analysts' coverage universe. If it does not appear in the Important Disclosures section of this report, the certifying Research Analyst's coverage universe can be found on J.P. Morgan's Research website, <https://www.jpmorganmarkets.com>.

J.P. Morgan Equity Research Ratings Distribution, as of July 04, 2026

	Overweight (buy)	Neutral (hold)	Underweight (sell)
J.P. Morgan Global Equity Research Coverage*	53%	36%	12%
IB clients**	83%	80%	73%
JPMS Equity Research Coverage*	51%	37%	12%
IB clients**	95%	92%	87%

*Please note that the percentages may not add to 100% because of rounding.

**Percentage of subject companies within each of the "buy," "hold" and "sell" categories for which J.P. Morgan has provided investment banking services within the previous 12 months.

For purposes of FINRA ratings distribution rules only, our Overweight rating falls into a buy rating category; our Neutral rating falls into a hold rating category; and our Underweight rating falls into a sell rating category. Please note that stocks with an NR designation are not included in the table above. This information is current as of the end of the most recent calendar quarter.

Equity Valuation and Risks: For valuation methodology and risks associated with covered companies or price targets for covered companies, please see the most recent company-specific research report at <http://www.jpmorganmarkets.com>, contact the primary analyst or your J.P. Morgan representative, or email research.disclosure.inquiries@jpmorgan.com. For material information about the proprietary models used, please see the Summary of Financials in company-specific research reports and the Company Tearsheets, which are available to download on the company pages of our client website, <http://www.jpmorganmarkets.com>. This report also sets out within it the material underlying assumptions used.

History of Investment Recommendations:

A history of J.P. Morgan investment recommendations disseminated during the preceding 12 months can be accessed on the Research &

Commentary page of <http://www.jpmorganmarkets.com> where you can also search by analyst name, sector or financial instrument.

Analysts' Compensation: The research analysts responsible for the preparation of this report receive compensation based upon various factors, including the quality and accuracy of research, client feedback, competitive factors, and overall firm revenues.

Registration of non-US Analysts: Unless otherwise noted, the non-US analysts listed on the front of this report are employees of non-US affiliates of J.P. Morgan Securities LLC, may not be registered as research analysts under FINRA rules, may not be associated persons of J.P. Morgan Securities LLC, and may not be subject to FINRA Rule 2241 or 2242 restrictions on communications with covered companies, public appearances, and trading securities held by a research analyst account.

Other Disclosures

J.P. Morgan is a marketing name for investment banking businesses of JPMorgan Chase & Co. and its subsidiaries and affiliates worldwide.

UK MIFID FICC research unbundling exemption: UK clients should refer to [UK MIFID Research Unbundling exemption](#) for details of J.P. Morgan's implementation of the FICC research exemption and guidance on relevant FICC research categorisation.

All research material made available to clients are simultaneously available on our client website, J.P. Morgan Markets, unless specifically permitted by relevant laws. Not all research content is redistributed, e-mailed or made available to third-party aggregators. For all research material available on a particular stock, please contact your sales representative.

Any long form nomenclature for references to China; Hong Kong; Taiwan; and Macau within this research material are Mainland China; Hong Kong SAR (China); Taiwan (China); and Macau SAR (China).

J.P. Morgan Research may, from time to time, write on issuers or securities targeted by economic or financial sanctions imposed or administered by the governmental authorities of the U.S., EU, UK or other relevant jurisdictions (Sanctioned Securities). Nothing in this report is intended to be read or construed as encouraging, facilitating, promoting or otherwise approving investment or dealing in such Sanctioned Securities. Clients should be aware of their own legal and compliance obligations when making investment decisions.

Any digital or crypto assets discussed in this research report are subject to a rapidly changing regulatory landscape. For relevant regulatory advisories on crypto assets, including bitcoin and ether, please see <https://www.jpmorgan.com/disclosures/cryptoasset-disclosure>.

The author(s) of this research report may not be licensed to carry on regulated activities in your jurisdiction and, if not licensed, do not hold themselves out as being able to do so.

Exchange-Traded Funds (ETFs): J.P. Morgan Securities LLC ("JPMS") acts as authorized participant for substantially all U.S.-listed ETFs. To the extent that any ETFs are mentioned in this report, JPMS may earn commissions and transaction-based compensation in connection with the distribution of those ETF shares and may earn fees for performing other trade-related services, such as securities lending to short sellers of the ETF shares. JPMS may also perform services for the ETFs themselves, including acting as a broker or dealer to the ETFs. In addition, affiliates of JPMS may perform services for the ETFs, including trust, custodial, administration, lending, index calculation and/or maintenance and other services.

Options and Futures related research: If the information contained herein regards options- or futures-related research, such information is available only to persons who have received the proper options or futures risk disclosure documents. Please contact your J.P. Morgan Representative or visit <https://www.theocc.com/components/docs/riskstoc.pdf> for a copy of the Option Clearing Corporation's Characteristics and Risks of Standardized Options or https://www.finra.org/sites/default/files/2020-08/Security_Futures_Risk_Disclosure_Statement_2020.pdf for a copy of the Security Futures Risk Disclosure Statement.

Changes to Interbank Offered Rates (IBORs) and other benchmark rates: Certain interest rate benchmarks are, or may in the future become, subject to ongoing international, national and other regulatory guidance, reform and proposals for reform. For more information, please consult: https://www.jpmorgan.com/global/disclosures/interbank_offered_rates

Private Bank Clients: Where you are receiving research as a client of the private banking businesses offered by JPMorgan Chase & Co. and its subsidiaries ("J.P. Morgan Private Bank"), research is provided to you by J.P. Morgan Private Bank and not by any other division of J.P. Morgan, including, but not limited to, the J.P. Morgan Corporate and Investment Bank and its Global Research division.

Legal entity responsible for the production and distribution of research: The legal entity identified below the name of the Reg AC Research Analyst who authored this material is the legal entity responsible for the production of this research. Where multiple Reg AC Research Analysts authored this material with different legal entities identified below their names, these legal entities are jointly responsible for the production of this research. Where more than one legal entity is listed under an analyst's name, the first legal entity is responsible for the production unless stated otherwise. Research Analysts from various J.P. Morgan affiliates may have contributed to the production of this material but may not be licensed to carry out regulated activities in your jurisdiction (and do not hold themselves out as being able to do so). Unless otherwise stated below in the legal entity disclosures, this material has been distributed by the legal entity responsible for production, or where more than one legal entity is listed under the analyst's name, the first legal entity will be responsible for distribution. If you have any queries, please contact the relevant Research Analyst in your jurisdiction or the entity in your jurisdiction that has distributed this research material.

Legal Entities Disclosures and Country-/Region-Specific Disclosures:

Argentina: JPMorgan Chase Bank N.A Sucursal Buenos Aires is regulated by Banco Central de la República Argentina ("BCRA"- Central Bank of Argentina) and Comisión Nacional de Valores ("CNV"- Argentinian Securities Commission - ALYC y AN Integral N°51).

Australia: J.P. Morgan Securities Australia Limited ("JPMSAL") (ABN 61 003 245 234/AFS Licence No: 238066) is regulated by the Australian Securities and Investments Commission and is a Market Participant of ASX Limited, a Clearing and Settlement Participant of ASX Clear Pty Limited and a Clearing Participant of ASX Clear (Futures) Pty Limited. This material is issued and distributed in Australia by or on behalf of JPMSAL only to "wholesale clients" (as defined in section 761G of the Corporations Act 2001). A list of all financial products covered can be found by visiting <https://www.jpmm.com/research/disclosures>. J.P. Morgan seeks to cover companies of relevance to the domestic and international investor base across all Global Industry Classification Standard (GICS) sectors, as well as across a range of market capitalisation sizes. If applicable, in the course of conducting public side due diligence on the subject company(ies), the Research Analyst team may at times perform such diligence through corporate engagements such as site visits, discussions with company representatives, management presentations, etc. Research issued by JPMSAL has been prepared in accordance with J.P. Morgan Australia's Research Independence Policy which can be found at the following link: [J.P. Morgan Australia - Research Independence Policy](#).

Brazil: Banco J.P. Morgan S.A. is regulated by the Comissao de Valores Mobiliarios (CVM) and by the Central Bank of Brazil. Ombudsman J.P. Morgan: 0800-7700847 / 0800-7700810 (For Hearing Impaired) / ouvidoria.jp.morgan@jpmchase.com.

Canada: J.P. Morgan Securities Canada Inc. is a registered investment dealer, regulated by the Canadian Investment Regulatory Organization and the Ontario Securities Commission and is the participating member on Canadian exchanges. This material is distributed in Canada by or on behalf of J.P.Morgan Securities Canada Inc.

Chile: Inversiones J.P. Morgan Limitada is an unregulated entity incorporated in Chile.

China: J.P. Morgan Securities (China) Company Limited has been approved by CSRC to conduct the securities investment consultancy business.

Colombia: Banco J.P. Morgan Colombia S.A. is supervised by the Superintendencia Financiera de Colombia (SFC). Any reference in this material to products or services offered abroad by entities other than the Bank in Colombia is included exclusively for descriptive purposes. Such references do not constitute, and should not be construed as, promotional activity or the provision of financial products or services within Colombian territory, as defined under applicable Colombian regulation.

Dubai International Financial Centre (DIFC): JPMorgan Chase Bank, N.A., Dubai Branch is regulated by the Dubai Financial Services Authority (DFSA) and its registered address is Dubai International Financial Centre - The Gate, West Wing, Level 3 and 9 PO Box 506551, Dubai, UAE. This material has been distributed by JP Morgan Chase Bank, N.A., Dubai Branch to persons regarded as professional clients or market counterparties as defined under the DFSA rules.

European Economic Area (EEA): Unless specified to the contrary, research is distributed in the EEA by J.P. Morgan SE ("JPM SE"), which is authorised as a credit institution by the Federal Financial Supervisory Authority (Bundesanstalt für Finanzdienstleistungsaufsicht, BaFin) and jointly supervised by the BaFin, the German Central Bank (Deutsche Bundesbank) and the European Central Bank (ECB). JPM SE is a company headquartered in Frankfurt with registered address at TaunusTurm, Taunustor 1, Frankfurt am Main, 60310, Germany. The material has been distributed in the EEA to persons regarded as professional investors (or equivalent) pursuant to Art. 4 para. 1 no. 10 and Annex II of MiFID II and its respective implementation in their home jurisdictions ("EEA professional investors"). This material must not be acted on or relied on by persons who are not EEA professional investors. Any investment or investment activity to which this material relates is only available to EEA relevant persons and will be engaged in only with EEA relevant persons.

Hong Kong: J.P. Morgan Securities (Asia Pacific) Limited (CE number AAJ321) is regulated by the Hong Kong Monetary Authority and the Securities and Futures Commission in Hong Kong, and J.P. Morgan Broking (Hong Kong) Limited (CE number AAB027) is regulated by the Securities and Futures Commission in Hong Kong. JP Morgan Chase Bank, N.A., Hong Kong Branch (CE Number AAL996) is regulated by the Hong Kong Monetary Authority and the Securities and Futures Commission, is organized under the laws of the United States with limited liability. Where the distribution of this material is a regulated activity in Hong Kong, the material is distributed in Hong Kong by or through J.P. Morgan Securities (Asia Pacific) Limited and/or J.P. Morgan Broking (Hong Kong) Limited.

India: J.P. Morgan India Private Limited (Corporate Identity Number - U67120MH1992FTC068724), having its registered office at J.P. Morgan Tower, Off. C.S.T. Road, Kalina, Santacruz - East, Mumbai - 400098, is registered with the Securities and Exchange Board of India (SEBI) as a 'Research Analyst' having registration number INH000001873. J.P. Morgan India Private Limited is also registered with SEBI as a member of the National Stock Exchange of India Limited and the Bombay Stock Exchange Limited (SEBI Registration Number - INZ000239730) and as a Merchant Banker (SEBI Registration Number - MB/INM000002970). Telephone: 91-22-6157 3000, Facsimile: 91-22-6157 3990 and Website: <http://www.jpmpi.com>. JPMorgan Chase Bank, N.A. - Mumbai Branch is licensed by the Reserve Bank of India (RBI) (Licence No. 53/ Licence No. BY.4/94; SEBI - IN/CUS/014/ CDSL : IN-DP-CDSL-444-2008/ IN-DP-NSDL-285-2008/ INBI00000984/ INE231311239) as a Scheduled Commercial Bank in India, which is its primary license allowing it to carry on Banking business in India and other activities, which a Bank branch in India are permitted to undertake. For non-local research material, this material is not distributed in India by J.P. Morgan India Private Limited. Compliance Officer: Ashutosh Sharma; ashutosh.j.sharma@jpmchase.com; +912261575002. Grievance Officer: Ramprasad K, jpmpi.research.feedback@jpmorgan.com; +912261573000. Registration granted by SEBI and certification from NISM in no way guarantee performance of the intermediary or provide any assurance of returns to investors. Please visit [Terms and Conditions and Most Important Terms and Conditions \(MITC\)](#). The annual Compliance audit report is available at <http://www.jpmpi.com/#research>.

Indonesia: PT J.P. Morgan Sekuritas Indonesia is a member of the Indonesia Stock Exchange and is registered and supervised by the Otoritas Jasa Keuangan (OJK).

Korea: J.P. Morgan Securities (Far East) Limited, Seoul Branch, is a member of the Korea Exchange (KRX). JPMorgan Chase Bank, N.A., Seoul Branch, is licensed as a branch office of foreign bank (JPMorgan Chase Bank, N.A.) in Korea. Both entities are regulated by the Financial Services Commission (FSC) and the Financial Supervisory Service (FSS). For non-macro research material, the material is distributed in Korea by or through J.P. Morgan Securities (Far East) Limited, Seoul Branch.

Japan: JPMorgan Securities Japan Co., Ltd. and JPMorgan Chase Bank, N.A., Tokyo Branch are regulated by the Financial Services Agency in Japan.

Malaysia: This material is issued and distributed in Malaysia by JPMorgan Securities (Malaysia) Sdn Bhd (18146-X), which is a Participating Organization of Bursa Malaysia Berhad and holds a Capital Markets Services License issued by the Securities Commission in Malaysia.

Mexico: J.P. Morgan Casa de Bolsa, S.A. de C.V., J.P. Morgan Grupo Financiero is member of the Mexican Stock Exchange ("Bolsa Mexicana de Valores") and the Institutional Stock Exchange ("Bolsa Institucional de Valores"), and it is authorized to act as a broker dealer by the National Banking and Securities Exchange Commission ("Comisión Nacional Bancaria y de Valores").

New Zealand: This material is issued and distributed by JPMSAL in New Zealand only to "wholesale clients" (as defined in the Financial Markets Conduct Act 2013). JPMSAL is registered as a Financial Service Provider under the Financial Service providers (Registration and Dispute Resolution) Act of 2008.

Philippines: J.P. Morgan Securities Philippines Inc. is a Trading Participant of the Philippine Stock Exchange and a member of the Securities Clearing Corporation of the Philippines and the Securities Investor Protection Fund. It is regulated by the Securities and Exchange Commission.

Singapore: This material is issued and distributed in Singapore by or through J.P. Morgan Securities Singapore Private Limited (JPMS) [MDDI (P) 057/08/2025 and Co. Reg. No.: 199405335R], which is a member of the Singapore Exchange Securities Trading Limited, and/or JPMorgan Chase Bank, N.A., Singapore branch (JPMCB Singapore), both of which are regulated by the Monetary Authority of Singapore. This material is issued and distributed in Singapore only to accredited investors, expert investors and institutional investors, as defined in Section 4A of the Securities and Futures Act, Cap. 289 (SFA). This material is not intended to be issued or distributed to any retail investors or any other investors that do not fall into the classes of "accredited investors," "expert investors" or "institutional investors," as defined under Section 4A of the SFA. Recipients of this material in Singapore are to contact JPMS or JPMCB Singapore in respect of any matters arising from, or in connection with, the material.

South Africa: J.P. Morgan Equities South Africa Proprietary Limited and JPMorgan Chase Bank, N.A., Johannesburg Branch are members of the Johannesburg Securities Exchange and are regulated by the Financial Services Conduct Authority (FSCA).

Taiwan: J.P. Morgan Securities (Taiwan) Limited is a participant of the Taiwan Stock Exchange (company-type) and regulated by the Taiwan Securities and Futures Bureau. Material relating to equity securities is issued and distributed in Taiwan by J.P. Morgan Securities (Taiwan) Limited, subject to the license scope and the applicable laws and the regulations in Taiwan. **To the extent that J.P. Morgan Securities (Taiwan) Limited produces research materials on securities not listed on the Taiwan Stock Exchange or Taipei Exchange ("Non-Taiwan Listed Securities"), these materials shall not constitute securities recommendations for the purpose of applicable Taiwan regulations, and, for the avoidance of doubt, J.P. Morgan Securities (Taiwan) Limited does not act as broker for Non-Taiwan Listed Securities.** According to Paragraph 2, Article 7-1 of Operational Regulations Governing Securities Firms Recommending Trades in Securities to Customers (as amended or supplemented) and/or other applicable laws or regulations, please note that the recipient of this material is not permitted to engage in any activities in connection with the material that may give rise to conflicts of interests, unless otherwise disclosed in the "Important Disclosures" in this material.

Thailand: This material is issued and distributed in Thailand by JPMorgan Securities (Thailand) Ltd., which is a member of the Stock Exchange of Thailand and is regulated by the Ministry of Finance and the Securities and Exchange Commission. The registered address is 548 One City Center Building, 50th Floor, Ploenchit Road, Lumpini, Pathum Wan, Bangkok 10330.

UK: Research is produced in the UK by J.P. Morgan Securities plc ("JPMS plc") which is a member of the London Stock Exchange and is authorised by the Prudential Regulation Authority and regulated by the Financial Conduct Authority and the Prudential Regulation Authority or J.P. Morgan Markets Limited ("JPML Ltd") which is authorised and regulated by the Financial Conduct Authority. Unless specified to the contrary, this material is distributed in the UK by JPMS plc and is directed in the UK only to: (a) persons having professional experience in matters relating to investments falling within article 19(5) of the Financial Services and Markets Act 2000 (Financial Promotion) (Order) 2005 ("the FPO"); (b) persons outlined in article 49 of the FPO (high net worth companies, unincorporated associations or partnerships, the trustees of high value trusts, etc.); or (c) any persons to whom this communication may otherwise lawfully be made; all such persons being referred to as "UK relevant persons". This material must not be acted on or relied on by persons who are not UK relevant persons. Any investment or investment activity to which this material relates is only available to UK relevant persons and will be engaged in only with UK relevant persons. A description of J.P. Morgan EMEA's policy for prevention and avoidance of conflicts of interest related to the production of Research can be found at the following link: [J.P. Morgan EMEA - Research Independence Policy](#).

U.S.: J.P. Morgan Securities LLC ("JPMS") is a member of the NYSE, FINRA, SIPC, and the NFA. JPMorgan Chase Bank, N.A. is a member of the FDIC. Material published by non-U.S. affiliates is distributed in the U.S. by JPMS who accepts responsibility for its content.

General: Additional information is available upon request. The information in this material has been obtained from sources believed to be reliable. While all reasonable care has been taken to ensure that the facts stated in this material are accurate and that the forecasts, opinions and expectations contained herein are fair and reasonable, JPMorgan Chase & Co. or its affiliates and/or subsidiaries (collectively J.P. Morgan) make no representations or warranties whatsoever to the completeness or accuracy of the material provided, except with respect to any disclosures relative to J.P. Morgan and the Research Analyst's involvement with the issuer that is the subject of the material. Accordingly, no reliance should be placed on the accuracy, fairness or completeness of the information contained in this material. There may be certain discrepancies with data and/or limited content in this material as a result of calculations, adjustments, translations to different languages, and/or local regulatory restrictions, as applicable. These discrepancies should not impact the overall investment analysis, views and/or recommendations of the subject company(ies) that may be discussed in the material. Artificial intelligence tools may have been used in the preparation of this material, including assisting in data analysis, pattern recognition, and content drafting for research material. J.P. Morgan accepts no liability whatsoever for any loss arising from any use of this material or its contents, and neither J.P. Morgan nor any of its respective directors, officers or employees, shall be in any way responsible for the contents hereof, apart from the liabilities and responsibilities that may be imposed on them by the relevant regulatory authority in the jurisdiction in question, or the regulatory regime thereunder. Opinions, forecasts or projections contained in this material represent J.P. Morgan's current opinions or judgment as of the date of the material only and are therefore subject to change without notice. Periodic updates may be provided on companies/industries based on company-specific developments or announcements, market conditions or any other publicly available information. There can be no assurance that future results or events will be consistent with any such opinions, forecasts or projections, which represent only one possible outcome. Furthermore, such opinions, forecasts or projections are subject to certain risks, uncertainties and assumptions that have not been verified, and future actual results or events could differ materially. The value of, or income from, any investments referred to in this material may fluctuate and/or be affected by changes in exchange rates. All pricing is indicative as of the close of market for the securities discussed, unless otherwise stated. Past performance is not indicative of future results. Accordingly, investors may receive back less than originally invested. This material is not intended as an offer or solicitation for the purchase or sale of any financial instrument. The opinions and recommendations herein do not take into account individual client circumstances, objectives, or needs and are not intended as recommendations of particular securities, financial instruments or strategies to particular clients. This material may include views on structured securities, options, futures and other derivatives. These are complex instruments, may involve a high degree of risk and may be appropriate investments only for sophisticated investors who are capable of understanding and assuming the risks involved. The recipients of this material must make their own independent decisions regarding any securities or financial instruments mentioned herein and should seek advice from such independent financial, legal, tax or other adviser as they deem necessary. J.P. Morgan may trade as a principal on the basis of the Research Analysts' views and research, and it may also engage in transactions for its own account or for its clients' accounts in a manner inconsistent with the views taken in this material, and J.P. Morgan is under no obligation to ensure that such other communication is brought to the attention of any recipient of this material. Others within J.P. Morgan, including Strategists, Sales staff and other Research Analysts, may take views that are inconsistent with those taken in this material. Employees of J.P. Morgan not involved in the preparation of this material may have investments in the securities (or derivatives of such securities) mentioned in this material and may trade them in ways different from those discussed in this material. This material is not an advertisement for or marketing of any issuer, its products or services, or its securities in any jurisdiction.

Confidentiality and Security Notice: This transmission may contain information that is privileged, confidential, legally privileged, and/or exempt from disclosure under applicable law. If you are not the intended recipient, you are hereby notified that any disclosure, copying, distribution, or use of the information contained herein (including any reliance thereon) is STRICTLY PROHIBITED. Although this transmission and any attachments are believed to be free of any virus or other defect that might affect any computer system into which it is received and opened, it is the responsibility of the recipient to ensure that it is virus free and no responsibility is accepted by JPMorgan Chase & Co., its subsidiaries and affiliates, as applicable, for any loss or damage arising in any way from its use. If you received this transmission in error, please immediately contact the sender and destroy the material in its entirety, whether in electronic or hard copy format. This message is subject to electronic monitoring: <https://www.jpmorgan.com/disclosures/email>

MSCI: Certain information herein ("Information") is reproduced by permission of MSCI Inc., its affiliates and information providers ("MSCI") ©2026. No reproduction or dissemination of the Information is permitted without an appropriate license. MSCI MAKES NO EXPRESS OR IMPLIED WARRANTIES (INCLUDING MERCHANTABILITY OR FITNESS) AS TO THE INFORMATION AND DISCLAIMS ALL LIABILITY TO THE EXTENT PERMITTED BY LAW. No Information constitutes investment advice, except for any applicable Information from MSCI ESG Research. Subject also to [msci.com/disclaimer](https://www.msci.com/disclaimer)

Sustainalytics: Certain information, data, analyses and opinions contained herein are reproduced by permission of Sustainalytics and: (1) includes the proprietary information of Sustainalytics; (2) may not be copied or redistributed except as specifically authorized; (3) do not constitute investment advice nor an endorsement of any product or project; (4) are provided solely for informational purposes; and (5) are not warranted to be complete, accurate or timely. Sustainalytics is not responsible for any trading decisions, damages or other losses related to it or its use. The use of the data is subject to conditions available at <https://www.sustainalytics.com/legal-disclaimers>. ©2026 Sustainalytics. All Rights Reserved.

"Other Disclosures" last revised July 04, 2026.

Copyright 2026 JPMorgan Chase & Co. All rights reserved. This material or any portion hereof may not be reprinted, sold or redistributed without the written consent of J.P. Morgan. It is strictly prohibited to use or share without prior written consent from J.P. Morgan any research material received from J.P. Morgan or an authorized third-party ("J.P. Morgan Data") in any third-party artificial intelligence ("AI") systems or models when such J.P. Morgan Data is accessible by a third-party.

Completed 29 Jul 2026 10:11 PM EDT

Disseminated 29 Jul 2026 10:30 PM EDT